

IDENTIFICATION

PRODUCT CODE: MAINDEC-DB-DHTMS-A-D

PRODUCT NAME: TMS-E DRIVE FUNCTION TIMER

DATE CREATED: JULY 16, 1973

MAINTAINER: DIAGNOSTIC GROUP

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NOTE

THERE ARE SIX DIAGNOSTIC PROGRAMS ASSOCIATED WITH THE TMB-E DECMAGTAPE CONTROL AND ITS TRANSPORT SYSTEM, ALTHOUGH PHYSICALLY SEPARATE, THESE PROGRAMS MUST BE TREATED AS A LARGE INTEGRATED TEST, AND TO ENSURE PROPER SYSTEM OPERATION, THESE TESTS MUST BE EXECUTED IN THE ORDER DELINEATED BELOW.

IF A GIVEN TEST SHOULD FAIL AND IT APPEARS THAT A FIX HAS BEEN FOUND, ALL PROGRAMS MUST ONCE AGAIN BE RUN, ONLY WHEN ALL TESTS HAVE RUN WITHOUT ANY UNACCEPTABLE ERRORS CAN THE TMB-E SYSTEM BE CONSIDERED UP.

TMB-E DIAGNOSTIC PROGRAMS' ORDER OF EXECUTION

1. TMB-E CONTROL TEST PART 1 (MAINDEC-08-DHTMA)
2. TMB-E CONTROL TEST PART 2 (MAINDEC-08-DHTMB)
3. TMB-E DRIVE FUNCTION TIMER (MAINDEC-08-DHTMC)
4. TMB-E DATA RELIABILITY 9 TRACK (MAINDEC-08-DHTMD)
5. TMB-E DATA RELIABILITY 7 TRACK (MAINDEC-08-DHTME)
6. TMB-E RANDOM EXERCISER (MAINDEC-08-DHTMF)

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1. ABSTRACT

THE TM8E DRIVE FUNCTION TIMER ASSISTS IN THE TESTING OF THE TM8E CONTROL UNIT AND TU10 TAPE UNIT. SELECTED OPERATIONS ARE EXECUTED, TIMED, AND THE TIMES ARE THEN PRINTED (IN MILLISECONDS). THERE IS NO LIMIT OR ERROR TESTING FACILITIES IN THE PROGRAM, THE DECISION ON THE VALIDITY OF TIMES MEASURED MUST BE MADE BY THE OPERATOR. ANY CONFIGURATION OF UP TO 8 TU10 TAPE UNITS (7 AND 9 CHANNEL) MAY BE SELECTED.

2. REQUIREMENTS

2.1 HARDWARE

PDP-8/E, 8/M, 8/F
TELETYPE OR COMPATIBLE DEVICE (TTY)
TM8-E DECMAGTAPE CONTROL
TU10 MASTER/SLAVE TRANSPORT SYSTEM

2.2 MEMORY

THIS PROGRAM REQUIRES 4K OF MEMORY AND MAY RESIDE IN ANY MEMORY FIELD.

2.3 PRELIMINARY PROGRAMS

ALL PROCESSOR/MEMORY DIAGNOSTICS
TM8-E CONTROL TEST PART 1
TM8-E CONTROL TEST PART 2

3. PROGRAM LOADING PROCEDURE

LOAD THE PROGRAM INTO ANY DESIRED MEMORY FIELD USING THE STANDARD BINARY LOADER TECHNIQUE.

4. PROGRAM STARTING PROCEDURE

- A. LOAD ADDRESS 0200
- B. LOAD THE EXTENDED ADDRESS WITH THE PROGRAM FIELD.
- C. CLEAR SWITCHES
- D. DEPRESS CLEAR, THEN CONTINUE

E. THE PROGRAM WILL PRINT ITS TITLE AND MAINDEC NUMBER, THEN ASK FOR DRIVE SELECTION, PRIOR TO MAKING DRIVE SELECTION, GO TO THE STANDARD TEST PROCEDURE IN PARAGRAPH 5.

NOTE! THE PROGRAM MAY BE RESTARTED AT ANY TIME AT ADDRESS 0201. IN THIS CASE THE PROGRAM BYPASSES PRINTING ITS TITLE AND IMMEDIATELY ASKS FOR DRIVE SELECTION.

5. STANDARD TEST PROCEDURE -----

USE OF THE STANDARD TEST PROCEDURE ENSURES PROPER TMB-E/TU10 CHECKOUT. INFORMATION REGARDING THE TIME LIMIT SPECIFICATIONS FOR THE TAPE OPERATIONS TIMED BY THIS PROGRAM ARE GIVEN IN PARAGRAPH 7.1. NO ERROR DETECTION OR DECISIONS REGARDING PROPER FUNCTION TIMES ARE CARRIED OUT BY THE PROGRAM.

5.1 DRIVE SELECTION -----

TO SPECIFY THE DRIVE(S) TO BE TESTED, CARRY OUT THE FOLLOWING STEPS:

- A. MOUNT A SPARE REEL OF INDUSTRY COMPATIBLE MAGNETIC TAPE ON ALL DRIVES TO BE SELECTED WITH THE WRITE ENABLE RING IN PLACE (WRITE ENABLED). LOAD THE TAPES AND POSITION AT BOT. ASSIGN DRIVE NUMBERS IN THE FOLLOWING SEQUENCE! 01711;6121513;41 AND THEN SWITCH ON LINE.
- B. START OR RESTART THE PROGRAM AS DESCRIBED IN PARAGRAPH 4.
- C. RESPOND TO "SELECT DRIVES" BY SETTING SR BIT N = 1 FOR EACH DRIVE TO BE TESTED (DRIVE N). THEN DEPRESS CONTINUE.
- D. RESPOND TO "SELECT 7 AND/OR 9 TRACK OPERATION" BY SETTING SR BIT N = 1 ONLY FOR THOSE SELECTED DRIVES WHICH ARE 9 TRACK UNITS. THEN DEPRESS CONTINUE.

5.2 TEST PROCEDURE -----

ACCOMPLISH THE FOLLOWING STEPS FOR ALL DRIVES IN THE SYSTEM UNDER TEST.

- A. SELECT ALL DRIVES IN THE SYSTEM UNDER TEST AS DESCRIBED IN PARAGRAPH 5.1 ABOVE.
- B. AS SOON AS TRACK SELECTION IS COMPLETE, THE PROGRAM WILL START TIMING THE VARIOUS TAPE OPERATIONS AND INDICATING ALL TIMES ON THE TTY.
- C. REPEAT THE ENTIRE TIMING PROCEDURE 2 TIMES.
- D. COMPARE THE RESULTS AS PRINTED ON THE TTY WITH THE LIMITS SPECIFIED IN PARAGRAPH 7.1. ADDITIONAL INFORMATION IS

GIVEN IN THE PROGRAM DESCRIPTION IN PARAGRAPH 10.

6. PROGRAM CONTROLS

THERE ARE NO SWITCH REGISTER OPTIONS ASSOCIATED WITH THIS PROGRAM.

7. ERRORS

THERE IS NO ERROR DETECTION INCLUDED IN THIS PROGRAM. ALL DECISIONS MUST BE MADE BY THE USER BY COMPARING THE FUNCTION TIMES PRINTED ON THE TTY WITH THOSE TIMES GIVEN IN THE TABLE BELOW.

7.1 TIME LIMITS SPECIFICATION

THE TABLE BELOW LISTS THE TIME LIMITS IN THE SAME FORMAT AS THEY ARE PRINTED ON THE TTY. TIMES LISTED UNDER UNIT 0 ARE 9 TRACK TIMES, AND THOSE UNDER UNIT 1 ARE 7 TRACK TIMES. ALL TIMES ARE IN MILLISECONDS.

FUNCTION	(9TRACK)	+/-	(7TRACK)	+/-
800 BPI	UNIT0		UNIT1	
WR FM BOT DELAY	185	15	185	15
WRITE SHUTDOWN	2.7	0.5	2.7	0.5
WRITE START	8.9	1.0	12.6	1.0
WR NONSTOP GAP	11.5	2.0	14.5	2.0
BKSP SHOWN*SDWN	15	5	19	5
READ SHUTDOWN	1.8	0.5	1.8	0.5
WRITE XIRG	95	10	95	10
LAST CHR TO CUR	0.26	0.1	0.26	0.1
RD FM BOT DELAY	185	15	185	15
SPCE SHOWN*SDWN	14	5	14	5
WRITE EOF	100	10	105	10
ER TO EF SP TME	100	10	105	10
WR TO ERASE HEAD	12	5	12	5
1 INCH DATA TIME	22	1	22	1

GAP1

GAP2

GAP3

GAP4

GAPS! 8>7>6>5>4>1) 1-2<1.7) 2>3

GAP5

GAP6

GAP7

GAP8

556 BPI	UNIT1	+/- (7TRACK)
WR FM BOT DELAY	185	15
WRITE SHUTDOWN	2.7	0.5
LAST CHR TO CUR	0.35	0.1
BKSP SHOWN*SDWN	19	5
READ SHUTDOWN	1.8	0.5
1 INCH DATA TIME	22	1

	UNIT 1	+/- (7TRACK)
200 BPI	165	15
WR FM BOT DELAY	2.7	0.5
WRITE SHUTDOWN	1.05	0.2
LAST CHR TO CUR	19	5
BKSP SHOWN+SDWN	1.8	0.5
READ SHUTDOWN	22	1
1 INCH DATA TIME		

8. RESTRICTIONS

NONE.

9. EXECUTION TIME

EXECUTION TIME VARIES DEPENDING ON THE NUMBER AND TYPES OF DRIVES BEING TESTED.

10. PROGRAM DESCRIPTION

10.1 WRITE FROM BOT DELAY

THIS IS THE TIME NECESSARY TO MOVE BEGINNING OF TAPE (BOT) MARKER APPROXIMATELY 6 INCHES PAST THE WRITE HEAD. THE FIRST RECORD ON TAPE MUST BE WRITTEN AT LEAST 3" AWAY FROM THE BOT MARKER.

PROCEDURE TO MEASURE TIME:

- REWIND TO BOT.
- INITIALIZE WC & CA LOCATIONS.
- LOAD COMMAND REGISTER WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- LOAD FUNCTION WITH WRITE & GO BITS.
- NOMITOR CA REGISTER UNTIL IT INDICATES FIRST WORD HAS BEEN TAKEN.
- THE TIME FROM "GO" UNTIL FIRST WORD IS OUTPUT IS "WRITE FROM BOT DELAY."

10.2 WRITE SHUTDOWN

THIS IS THE TIME NECESSARY TO CONTINUE MOVING TAPE AFTER A RECORD IS WRITTEN SO THAT THE PROPER INTERRECORD GAP WILL EXIST BETWEEN RECORDS.

PROCEDURE TO MEASURE TIME:

- REWIND TO BOT.
- INITIALIZE WC & CA LOCATIONS TO WRITE ONE

200(10) WORD RECORD FROM BOT.

- C. LOAD COMMAND REGISTER WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- D. LOAD FUNCTION WITH WRITE & GO BITS.
- E. AFTER MTF SETS INDICATING THE END OF RECORD, INITIATE A CONTINUOUS WRITE AND TIME TILL THE FIRST WORD IS TRANSFERRED FROM MEMORY TO THE TMR-E CONTROL (OCCURS APPROXIMATELY ON THE LEADING EDGE OF SDWN).

10.3 WRITE START

THE TIME NECESSARY FOR TAPE TO ACCELERATE TO FULL SPEED + GUARANTEE A 1/2" INTERRECORD GAP.

PROCEDURE TO MEASURE TIME:

SAME AS "WRITE FROM BOT" EXCEPT NOW WE ARE NOT AT BOT.

- A. INITIALIZE WC & CA REGISTER.
- B. LOAD COMMAND WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- C. LOAD FUNCTION WITH WRITE & GO BITS.
- D. MONITOR CA FOR ITS SECOND INCREMENT.
- E. THE TIME FROM "GO" UNTIL 2ND INCREMENT OF CA IS APPROXIMATELY EQUAL TO "WRITE START".

10.4 WRITE NONSTOP GAP

WRITE NONSTOP GAP IS EQUIVALENT TO THE SUM OF "WRITE SHUTDOWN" & "WRITE START" AND IS THE TIME NECESSARY TO INSURE THAT THE INTERRECORD GAP WILL BE AT LEAST 1/2" WHEN WRITING NON-STOP.

PROCEDURE TO MEASURE TIME:

- A. INITIALIZE WC & CA LOCATIONS
- B. LOAD COMMAND WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- C. LOAD FUNCTION; WRITE, GO BITS.
- D. WAIT FOR "JOB DONE."
- E. ISSUE "CLF."
- F. RE-INITIALIZE WC & CA LOCATIONS.
- G. LOAD FUNCTION; WRITE, GO BITS.
- H. TIME FROM 2ND "GO" UNTIL FIRST WORD OF 2ND WRITE IS OUTPUT IS "WRITE NONSTOP GAP."

10.5 BACKSPACE SHUTDOWN + SETTLE DOWN DELAY

THIS IS THE LENGTH OF TIME NECESSARY TO GUARANTEE THAT IF A WRITE OPERATION FOLLOWS A BACKSPACE THE TAPE WILL BE POSITIONED SUCH THAT ALL PREVIOUS DATA IS IN FRONT OF THE WRITE AND ERASE HEADS AND WILL BE ERASED,

PROCEDURE TO MEASURE TIME

RECORD IS USED WHICH WAS JUST WRITTEN IN PREVIOUS TEST "WRITE NONSTOP GAP",

- A. LOAD COMMAND WITH DRIVE SELECT, 800 BPI, ODD PARITY,
- B. SET WC TO BACKSPACE 1 RECORD,
- C. ISSUE BACKSPACE, GO,
- D. AFTER JOB DONE BECOMES A ONE INDICATING THE BEGINNING OF THE RECORD MONITOR TUR UNTIL IT BECOMES A "1", THE TIME FROM JOB DONE UNTIL TUR IS "BACKSPACE SHUTDOWN + SETTLEDOWN DELAY,"

10.6 READ SHUTDOWN

THIS IS THE AMOUNT OF TIME NECESSARY TO CONTINUE MOVING TAPE AFTER A RECORD IS READ SO THAT THERE IS ENOUGH GAP FOR TAPE TO BE FULLY ACCELERATED IF A READ IS FOLLOWED BY A BACKSPACE, "READ SHUTDOWN" MUST ALSO BE LESS THAN "WRITE SHUTDOWN" TO GUARANTEE THAT THE WRITE AND ERASE HEADS WILL BE POSITIONED SUCH THAT ALL PREVIOUS DATA IS IN FRONT OF THE HEADS AND WILL BE ERASED IF A WRITE FOLLOWS A READ, IN ADDITION WHEN A WRITE FOLLOWS A READ THE INTERRECORD GAP MUST STILL BE AT LEAST 1/2",

PROCEDURE TO MEASURE TIME

- A. RECORD USED IN "BACKSPACE SHUTDOWN" IS READ,
- B. INITIALIZE WC & CA REGISTERS,
- C. ISSUE READ FUNCTION, 800 BPI, GO,
- D. AFTER MTF SETS INDICATING THE END OF RECORD, INITIATE A CONTINUOUS WRITE AND TIME TILL THE FIRST WORD IS TRANSFERRED FROM MEMORY TO THE TMS-E CONTROL (OCCURS APPROXIMATELY ON THE LEADING EDGE OF SDWN),

10.7 WRITE XIRG

THIS IS THE TIME NECESSARY TO CAUSE THE GENERATION OF AN INTERRECORD GAP THAT IS AT

LEAST 3" LONG AS COMPARED WITH THE NORMAL 3/8" GAP. THE PURPOSE IS TO ELIMINATE WRITE ERRORS THAT MAY BE CAUSED BY A DEFECTIVE AREA ON TAPE. NORMALLY ONE REWRITE WITH XIRG WOULD BE SUFFICIENT IF IT ISN'T THE PROCEDURE WOULD BE TO REPEAT THE "BACKSPACE-REWRITE WITH XIRG" SEQUENCE UNTIL A RECORD IS WRITTEN WITHOUT ERRORS. EACH SUCCESSIVE REWRITE WOULD ADD 3" TO THE INTERRECORD GAP UNTIL "GOOD" TAPE WAS REACHED.
PROCEDURE TO MEASURE TIME:

- A. TAPE IS NOT BOT.
- B. INITIALIZE WC & CA REGISTERS.
- C. ISSUE WRITE WITH XIRG FUNCTION, 800 BPI, GO.
- D. MONITOR CA TO DETERMINE WHEN FIRST WORD IS OUTPUT.
- E. THE TIME FROM "GO" UNTIL FIRST WORD IS OUTPUT IS "WRITE WITH XIRG".

10.8 LAST CHARACTER TO CU READY

THIS IS THE AMOUNT OF TIME IT TAKES FOR THE CONTROL TO SENSE 3 MISSING WORDS ON TAPE (END OF RECORD) UNTIL "CU READY" BECOMES A "1."

PROCEDURE TO MEASURE TIME:

- A. PROGRAM USES SAME RECORD THAT WAS WRITTEN DURING "WRITE XIRG."
- B. LOAD COMMAND WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- C. INITIALIZE WC TO SPACE REVERSE 1 RECORD.
- D. WAIT FOR JOB DONE.
- E. REINITIALIZE WC & CA LOCATIONS.
- F. LOAD COMMAND WITH DRIVE SELECT, 800 BPI, ODD PARITY.
- G. ISSUE READ FUNCTION, GO.
- H. WAIT UNTIL WC=0 AND THEN MONITOR "CU READY" UNTIL IT BECOMES A "1".
- J. THE TIME FROM WC=0 UNTIL "CU READY" = 1 IS "LAST CHARACTER TO CU READY".

10.9 READ FROM BOT

THE FIRST RECORD WRITTEN ON TAPE IS SUPPOSED

TO BE AT LEAST 6" FROM THE BOT MARKER.
IN THE EVENT THAT THIS CONDITION WASN'T MET
IT IS STILL DESIREABLE TO READ THE RECORD,
READ FROM BOT IS THE TIME FROM WHEN A READ
FUNCTION IS ISSUED UNTIL THE FIRST WORD IS
OUTPUT.

PROCEDURE TO MEASURE TIME:

- A. THE RECORD THAT WAS WRITTEN JUST OFF BOT
DURING "WRITE START" IS USED.
- B. TAPE IS REWOUND TO BOT.
- C. INITIALIZE WC & CA REGISTERS.
- D. ISSUE READ FUNCTION, 800 BPI, GO.
- E. MONITOR CA TO DETERMINE WHEN FIRST WORD
IS OUTPUT.
- F. THE TIME FROM "GO" UNTIL FIRST BREAK IS
"READ FROM BOT."

10.10 SPACE SHUTDOWN + SETTLEDOWN DELAY

SPACE SHUTDOWN IS THE AMOUNT OF TIME NECESSARY
TO CONTINUE MOVING TAPE AFTER A RECORD IS
SPACED OVER IN THE FORWARD DIRECTION FOR THE
SAME REASONS AS "READ SHUTDOWN".

PROCEDURE TO MEASURE TIME:

- A. SAME RECORD IS USED AS IN T3 WRITE START.
- B. INITIALIZE WC TO SPACE FORWARD 1 RECORD.
- C. ISSUE SPACE FORWARD FUNCTION, 800 BPI, GO.
- D. MONITOR JOB DONE UNTIL IT IS =1.
- E. TIME FROM JOB DONE =1 UNTIL TUR IS
"SPACE SHUTDOWN + SETTLEDOWN DELAY".

10.11 WRITE EOF

TO WRITE AN END OF FILE MARK IT IS NECESSARY
FOR TAPE TO MOVE 3 INCHES BEFORE WRITING
A RECORD WITH EXTENDED INTERRECORD GAP.
HOWEVER, AN EOF MARK CORRESPONDS TO A
1 WORD RECORD, THE TIME SHOULD BE
SLIGHTLY LARGER THAN "WRITE XIRG".

PROCEDURE TO MEASURE TIME:

- A. TAPE UNIT IS REWOUND TO BOT.
- B. INITIALIZE WC & CA REGISTERS.

- C. ISSUE WRITE, 800 BPI, GO.
- D. WAIT FOR JOB DONE.
- E. ISSUE CLF.
- F. ISSUE WRITE EOF FUNCTION, 800 BPI, GO.
- G. WAIT FOR JOB DONE =1
- H. THE TIME FROM "GO" UNTIL "JOB DONE" IS "WRITE EOF."

10.12 EOR TO EOF TIME

THIS IS THE TIME NEEDED TO MOVE TAPE FROM THE END OF A RECORD TO AN END OF FILE MARK WRITTEN AFTER IT. THE PROCEDURE USED TURNS OUT TO BE A TEST OF THE WRITE AND ERASE HEAD POLARITIES. IF THE TIME PRINTED IS EQUAL TO ZERO IT IS AN INDICATION THAT THE EOF WAS NOT FOUND WHEN "JOB DONE" BECAME A ONE.

THIS COULD INDICATE ONE OR MORE OF THE FOLLOWING PROBLEMS:

- 1. ERASE HEAD POLARITY REVERSED.
- 2. ERASE HEAD CURRENT NOT SUFFICIENT TO FULLY SATURATE TAPE.
- 3. ONE OR MORE OF WRITE HEAD TRACKS POLARITY REVERSED.
- 4. ONE OR MORE SENSITIVE READ AMPLIFIERS.
- 5. WRITE EOF FUNCTION DIDN'T REALLY WRITE AN EOF MARK.

OTHERWISE "EOR TO EOF SPACE TIME" SHOULD BE SLIGHTLY LARGER THAN "WRITE EOF."

PROCEDURE TO MEASURE TIME:

- A. A RECORD AND EOF WAS PREVIOUSLY WRITTEN FROM BOT FOR "WRITE EOF".
- B. TAPE IS REWOUND TO BOT.
- C. REWRITE RECORD OVER PREVIOUSLY WRITTEN RECORD.
- D. BACKSPACE OVER RECORD JUST WRITTEN.
- E. SET WC REGISTER TO SPACE 2 RECORDS.
- F. ISSUE SPACE FORWARD FUNCTION, SET GO.
- G. WAIT FOR WC REGISTER TO INDICATE THAT FIRST RECORD HAS BEEN SPACED OVER THEN MONITOR "CU READY" UNTIL IT BECOMES A

"1", AFTER "CU READY" CHECK TO SEE IF "EOF"
IS A 1 IN STATUS REGISTER, IF EOF IS NOT
SET THEN ZERO TIME COUNTER,

H. TIME FROM WC=1 UNTIL "CU READY" IS
"EOR TO EOF SPACE TIME,"

10.13 WRITE TO ERASE HEAD

THE "WRITE TO ERASE HEAD" TEST INSURES THAT THE
TAPE IN FRONT OF THE WRITE HEAD IS ERASED
DURING A WRITE OPERATION,

PROCEDURE TO MEASURE TIMEI

- A. A LONG RECORD HAS BEEN WRITTEN FROM BOT, SAME
RECORD THAT WAS USED TO TIME "WRITE FROM BOT
DELAY"
- B. TAPE IS REWOUND TO BOT,
- C. WC REGISTER IS INITIALIZED FOR A 3 WORD RECORD
CA REGISTER IS INITIALIZED TO 3775,
- D. LOAD COMMAND REGISTER WITH DRIVE SELECT,
800 BPI, ODD PARITY,
- E. LOAD FUNCTION REGISTER WITH WRITE & GO BITS,
- F. MONITOR CA UNTIL IT IS 3777 INDICATING 2 WORDS
HAVE BEEN OUTPUT, THEN ISSUE CLT WHICH
STOPS ALL DATA TRANSFERS & CAUSES THE DRIVE
TO SHUTDOWN,
- G. REWIND TO BOT,
- H. INITIALIZE CA TO 3775
- I. ISSUE READ FUNCTION, 800 BPI, SET GO,
- J. MONITOR CA UNTIL IT IS 3777 THEN
GO TO TIMING ROUTINE AND TIME UNTIL CA=4000
THIS TIME WILL INDICATE THE DISTANCE BETWEEN
THE LAST WORD OF THE NEW DATA AND THE
FIRST WORD OF THE OLD DATA WHICH IS ALSO
THE AMOUNT OF TAPE THAT WAS ERASED BY
THE ERASE HEAD DURING THE WRITE OPERATION,

10.14 ONE INCH DATA TIME

ONE INCH OF DATA, 800 BPI (ALSO 556 & 200 IF
7 CHANNEL UNIT), IS WRITTEN AND TIMED TO
DETERMINE IF TAPE IS MOVING AT PROPER SPEED,

PROCEDURE TO MEASURE TIMEI

- A. INITIALIZE WC & CA REGISTERS, CA(3776) WC(=2000)

- B. ISSUE WRITE FUNCTION, 800, 556 OR 200 BPI, GO.
- C. WAIT FOR CURRENT MEMORY ADDRESS REGISTER TO INDICATE FIRST WORD IS TAKEN AND THEN MONITOR WORD COUNT UNTIL EQUAL TO ZERO.
- D. TIME FROM FIRST WORD OUTPUT UNTIL WORD COUNT=0 IS "ONE INCH DATA TIME."

10.15 GAP CONSISTENCY

FOR PROPER OPERATION THE INTERRECORD GAPS ON TAPE MUST ALWAYS BE AT LEAST 1/2 OF AN INCH. THIS WILL ALLOW DATA WRITTEN USING ONE TAPE UNIT TO BE READ ON ANOTHER TAPE UNIT WHEN THE START/STOP CHARACTERISTICS OF EACH UNIT ARE DIFFERENT. THE MINIMUM GAP SIZE OF 1/2 AN INCH IS GENERATED WHEN A WRITE FOLLOWS A READ. ALL OTHER GAPS SHOULD BE LARGER DEPENDING ON HOW THEY WERE WRITTEN.

PROCEDURE TO MEASURE TIME:

- A. A TOTAL OF NINE RECORDS ARE WRITTEN ON TAPE (FROM BOT) UTILIZING DIFFERENT SEQUENCES TO GENERATE THE INTERRECORD GAPS.
- B. THE TAPE IS REWOUND TO BOT.
- C. INITIALIZE WORD COUNT AND CURRENT ADDRESS REGISTERS.
- D. ISSUE READ FUNCTION, 800 BPI, SET "GO".
- E. WAIT FOR "CU READY" TO BECOME A 1 THEN REPEAT STEP C AND RESET "GO" TO CONTINUE NONSTOP.
- F. MONITOR CA REGISTER TO DETERMINE WHEN FIRST WORD IS INPUT.
- G. THE TIME FROM WHEN "GO" IS RESET UNTIL THE FIRST WORD IS INPUT WILL REFLECT THE SIZE OF THE GAP.
- H. STEPS E,F ARE REPEATED UNTIL ALL 8 GAPS ARE MEASURED.

PROGRAM SEQUENCE FOR EACH GAP:

- GAP1 WRITE FOLLOWED BY A WRITE (NONSTOP).
- GAP2 WRITE FOLLOWED BY A WRITE (START/STOP).
- GAP3 READ FOLLOWED BY A WRITE (START/STOP).
- GAP4 WRITE=BACKSPACE FOLLOWED BY A WRITE (START/STOP).
- GAP5 SAME AS GAP4 EXCEPT WRITE=BACKSPACE REPEATED 2 TIMES.

GAP6 SAME AS GAP4 EXCEPT WRITE=BACKSPACE REPEATED 3 TIMES;
GAP7 SAME AS GAP4 EXCEPT WRITE=BACKSPACE REPEATED 4 TIMES;
GAP8 SAME AS GAP4 EXCEPT WRITE=BACKSPACE REPEATED 5 TIMES;
GAP LENGTHS SHOULD REFLECT THE FOLLOWING RELATIONSHIP:
8>7>6>5>4>1, 1-2<1,7, 2>3

10.16 FUNCTIONS AT 556 BPI

ALL OF THE PREVIOUS TESTS USED THE DENSITY OF 800 BPI.
IF A 7 CHANNEL DRIVE IS SELECTED IT IS USEFUL TO RUN SEVERAL OF
THE TESTS AGAIN USING DENSITY OF 556 BPI, REFERENCE
THE PROPER PARAGRAPHS FOR A DESCRIPTION OF EACH TEST.

10.17 FUNCTIONS AT 200 BPI.

SAME AS ABOVE,
REFERENCE 8.16, "FUNCTIONS AT 556 BPI".

11. LISTING (ATTACHED)

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0127 1000 K1000, 1000
0130 0212 K212, 0212
0131 0215 K215, 0215
0132 7777 KM1, -1
0133 7776 KM2, -2
0134 7775 KM3, -3
0135 7770 KM10, -10
0136 5616 XBAK, BAKSPA
0137 5600 XRITE1, RITE1
0140 5633 XREAD, READ
0141 5691 XREAD1, READ1
0142 0400 K0400, 0400
0143 0000 XBP1, 0
0144 2214 LINK16, T16
0145 5704 XEND, FINI
0146 5670 XREAD2, READ2
0147 0000 INCH, 0

0150 0260 NUMTBL, 260
0151 0261 261
0152 0262 262
0153 0263 263
0154 0264 264
0155 0265 265
0156 0266 266
0157 0267 267

0160 0000 ALLTBL, 0
0161 0000 0
0162 0000 0
0163 0000 0
0164 0000 0
0165 0000 0
0166 0000 0
0167 0000 0

0170 0000 TU7TBL, 0
0171 0000 0
0172 0000 0
0173 0000 0
0174 0000 0
0175 0000 0
0176 0000 0
0177 0000 0

0200 0200 *200
0201 7610 START, SKP CLA
0202 5206 RESTART, JMP SELECT
0203 4474 JMS I XMESS /BEGINNING OF READER.
0204 6000 MSG1
0205 4474 JMS I XMESS
0206 6017 MSG2

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//ROUTINE FOR TRANSPORT SELECTION AND
//7 OR 9 TRACK OPERATION,
//
0206 4474 SELECT, JMS I XMESS /SET SWITCHES TO SELECT TRANSPORTS.
0207 6033 MSG3
0210 7402 HLT
0211 7604 LAG /GET SWITCHES FOR TRANSPORT SELECTION.
0212 3344 DCA DRVSAV /SAVE SWITCHES.
0213 4474 JMS I XMESS /SET SWITCHES FOR 7 OR 9 TRACK OPERATION.
0214 6044 MSG4
0215 7402 HLT
0216 7604 LAG /GET SWITCHES FOR 7 OR 9 TRACK SELECTION.
0217 3343 DCA TRKSAV /SAVE SWITCHES.
0220 3060 DCA CNTALL /INITIALIZE COUNTERS.
0221 3061 DCA CNT7
0222 1377 TAD (DRV TAB) /SETUP POINTER
0223 3062 DCA DRVPNT /FOR DRIVE TABLE.
0224 1376 TAD (ALLTU) /SETUP POINTER FOR TABLE WHICH WILL HOLD
0225 3340 DCA XALL /ALL TU-10'S TO BE TESTED.
0226 1375 TAD (TU7) /SETUP POINTER FOR TABLE WHICH WILL HOLD ONLY
0227 3341 DCA XTU7 /7 TRACK TU-10'S TO BE TESTED.
0230 1374 TAD (NUMTBL)
0231 3052 DCA NUMPNT
0232 1373 TAD (ALLTBL)
0233 3053 DCA TBLALL
0234 1372 TAD (TU7TBL)
0235 3054 DCA TBLTU7
0236 1345 TAD K7770 /SETUP COUNTER TO EXIT THIS ROUTINE WHEN ALL
0237 3342 DCA DRVCNT /BITS OF SELECT WORDS HAVE BEEN CHECKED.
0240 1344 CHKWRD, TAD DRVSAV /PICKUP DRIVE SELECT WORD.
0241 7700 SMA CLA /MINUS AC INDICATES DRIVE IS SELECTED.
0242 5264 JMP UPDATE /THIS DRIVE NOT SELECTED UPDATE ALL POINTERS.
0243 1462 TAD I DRVPNT /SAVE DRIVE SELECTED IN TABLE CONTAINING
0244 3740 DCA I XALL /ALL DRIVES TO BE TESTED.
0245 1452 TAD I NUMPNT
0246 3453 DCA I TBLALL
0247 2060 ISR CNTALL /COUNTER FOR SELECT ALL DRIVES TABLE.
0250 1343 TAD TRKSAV /PICKUP TRACK SELECT WORD.
0251 7710 SPA CLA /MINUS AC INDICATES 9 TRACK OPERATION.
0252 5262 JMP UPDATE-2 /9 TRACK SELECTED, EXIT AND UPDATE POINTERS.
0253 1462 TAD I DRVPNT /7 TRACK SELECTED, SAVE IT IN 7 TRACK
0254 3741 DCA I XTU7 /SELECT TABLE.
0255 1452 TAD I NUMPNT
0256 3454 DCA I TBLTU7
0257 2061 ISR CNT7 /KEEP COUNT OF ALL 7 TRACK DRIVES SELECTED.

0260 2054 ISR TBLTU7 /UPDATE ALL TABLE POINTERS.
0261 2341 ISR XTU7
0262 2053 ISR TBLALL
0263 2340 ISR XALL
0264 2342 UPDATE, ISR DRVCNT /CHECKED ALL BITS OF SELECT WORDS?
0265 7410 SKP /NO, CONTINUE CHECKING
0266 5303 JMP ENDCCHK /YES, BEGIN TESTING SELECTED DRIVES.
0267 2062 ISR DRVPNT
0270 2052 ISR NUMPNT

```

```

0271 7188 CLL
0272 1344 TAD DRVSAV /SET DRIVE SELECT WORD;
0273 7804 RAL /SETUP NEXT BIT TO BE CHECKED,
0274 3344 DCA DRVSAV /SAVE IT.
0275 7188 CLL
0276 1344 TAD TRKSAV /SET TRACK SELECT WORD;
0277 7804 RAL /SETUP NEXT BIT TO BE CHECKED,
0278 3344 DCA TRKSAV /SAVE IT.
0301 7188 CLL
0302 5240 JMP CHKHRD /DO IT ALL AGAIN.
0303 1061 ENDCHK, TAD CNT7 /CHECK TO SEE IF ANY DRIVES
0304 7640 SEA CLA /HAVE BEEN SELECTED.
0305 5311 JMP ,+4
0306 1060 TAD CNTALL
0307 7650 SNA CLA
0310 5280 JMP SELECT /NO DRIVES SELECTED, RETURN TO START.
0311 1371 TAD (3767 /FILL WRITE BUFFER WITH ONES.
0312 3346 DCA ONES
0313 1378 TAD (-638
0314 3347 DCA ONESCT
0315 7848 CMA
0316 3090 DCA ONCE
0317 7840 CMA
0320 3746 DCA I ONES
0321 2346 ISR ONES
0322 2347 ISR ONESCT
0323 5317 JMP ,+4
0324 1367 TAD (6377
0325 3818 DCA INDX18
0326 4474 JMS I XMESS
0327 6067 MSUB5
0330 4474 JMS I XMESS
0331 6182 MSUB7
0332 4455 JMS I XHEAD
0333 4451 JMS I XCALF
0334 3985 DCA I NULL
0335 3110 DCA MILICT /REMO TIME COUNTERS,
0336 5737 JMP I LINK1 /BEGIN TESTING.
0337 0400 LINK1, T1

```

```

0340 0030 XALL, ALLTU
0341 0040 XTU7, TU7
0342 0000 DRVCNT, 0
0343 0000 TRKSAV, 0
0344 0000 DRVSAV, 0
0345 7778 K7778, 7778
0346 0000 ONES, 0
0347 0000 ONESCT, 0

```

```

0367 6377
0370 7188
0371 3767
0372 6178

```

```

0373 0160
0374 0190
0375 0048
0376 0032
0377 0020
0400

```

PAGE

//TIME WRITE FROM BOT DELAY//

//

```

0400 7300 T1, CLA CLL
0401 1377 TAD (OUT1 /SETUP T1 EXIT LOCATION.
0402 3071 DCA OUT
0403 4463 JMS I XROUND /REWIND ALL SELECTED DRIVES TO BOT.
0404 4465 RET1, JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
0405 3776 JMS I XWCCA /CA
0406 7472 -318 /NO
0407 4472 JMS I SELALL /SET DRIVE SELECT WORD;
0410 1806 TAD 000000 /COMBINE DRIVE SELECT, 888 SPI, ODD PARITY
0411 6785 LCMR /LOAD COMMAND.
0412 1113 TAD K4100
0413 6786 LFER /LOAD FUNCTION/ WRITE, GO BITS.
0414 4497 TIMER /TIME "BOT DELAY."
0415 6713 RGRAR /NOT TO BE USED BY TIMER.
0416 7718 SPA CLA /CONDITION TO REMAIN IN TIMING LOOP.
0417 6724 SKTR
0420 5217 JMP ,+1 /WAIT FOR PUR.
0421 6725 CLF
0422 5284 JMP RET1 /GO AGAIN ON NEXT DRIVE.
0423 4474 JMS I XMESS /MESSAGE ROUTINE
0424 6112 MSBT1
0425 1060 TAD CNTALL
0426 3184 DCA PRACT
0427 4475 JMS I XPRINT

```

//TIME WRITE SHUTDOWN//

//

```

0430 7300 T2, CLA CLL
0431 1376 TAD (OUT2 /SETUP T2 EXIT LOCATION.
0432 3071 DCA OUT
0433 4463 JMS I XROUND /REWIND ALL SELECTED DRIVES TO BOT.
0434 4465 RET2, JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
0435 3776 JMS I XWCCA /CA
0436 7475 -318 /NO
0437 4472 JMS I SELALL /SET DRIVE SELECT WORD;
0440 1806 TAD 000000 /COMBINE DRIVE SELECT, 888 SPI, ODD PARITY.
0441 6785 LCMR /LOAD COMMAND.
0442 1113 TAD K4100
0443 6786 LFER /LOAD FUNCTION/ WRITE, GO BITS.
0444 6723 SKJQ
0445 5244 JMP ,+1 /WAIT FOR JOB DONE.
0446 6725 CLF /CLEAR FLAG
0447 4775 JMS I XWCONT /CONTINUOUS WRITE
0448 5234 JMP RET2
0449 4474 JMS I XMESS /MESSAGE ROUTINE

```

21

```

0452 6123      MSGT2
0453 1060      TAO      CNTALL
0454 3104      DCA      PRNCT
0455 4475      JMS I    XPRINT
0456 5657      JMP I    LINK3
0457 0600      LINK3,   T3

0575 5400
0576 0451
0577 0423
0600          PAGE

//TIME WRITE START//
//
0600 7300      T3,      CLA CLL
0601 1377      TAO      (OUT3      /SETUP T3 EXIT LOCATION.
0602 3071      DCA      OUT        /
0603 4465      RET3,    JMS I    XWCCA /INITIALIZE WC + CA LOCATIONS.
0604 3776      3776      /CA
0605 7470      -310     /WC
0606 4472      JMS I    SELALL /GET DRIVE SELECT WORD.
0607 1066      TAO      ODD8000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY.
0610 6705      LCMR     /LOAD COMMAND.
0611 1113      TAO      K4100
0612 6706      LFGR     /LOAD FUNCTION/ WRITE, GO BITS.
0613 4437      TIMER    /TIME "WRITE START."
0614 6713      RCAR     /EXIT TIMER ON
0615 7710      SPA CLA  /CA=4000.
0616 6724      SKTR
0617 5216      JMP      .-1      /WAIT FOR TUR.
0620 6725      CLF
0621 5203      JMP      RET3     /DO AGAIN ON NEXT DRIVE.

0622 4474      OUT3,    JMS I    XMESS /MESSAGE ROUTINE
0623 6134      MSGT3
0624 1060      TAO      CNTALL
0625 3104      DCA      PRNCT
0626 4475      JMS I    XPRINT

//TIME WRITE NONSTOP GAP//
//
0627 7300      T4,      CLA CLL
0628 1376      TAO      (OUT4      /SETUP T4 EXIT LOCATION.
0629 3071      DCA      OUT        /
0632 4465      RET4,    JMS I    XWCCA /INITIALIZE WC + CA LOCATIONS.
0633 3776      3776      /CA
0634 7470      -310     /WC
0635 4472      JMS I    SELALL /GET DRIVE SELECT WORD.
0636 1066      TAO      ODD8000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY.
0637 6705      LCMR     /LOAD COMMAND.
0640 1113      TAO      K4100
0641 6706      LFGR     /LOAD FUNCTION/ WRITE, GO BITS.
0642 6723      SKJD
0643 5242      JMP      .-1      /WAIT FOR JOB DONE.
0644 6725      CLF

```

```

0645 4465      JMS I    XWCCA /RE-INITIALIZE WC + CA LOCATIONS.
0646 3776      3776      /CA
0647 7470      -310     /WC
0650 1375      TAO      (3      /PRESET TIME COUNTER WITH
0651 3505      DCA I    NULL /DURATION OF THIS SETUP PROCEDURE.
0652 1113      TAO      K4100
0653 6706      LFGR     /LOAD FUNCTION/ WRITE, GO BITS.
0654 4497      TIMER    /TIME "NONSTOP GAP."
0655 6713      RCAR     /EXIT TIMER ON
0656 7710      SPA CLA  /CA=4000.
0657 6723      SKJD
0660 5257      JMP      .-1      /WAIT FOR JOB DONE.
0661 6725      CLF
0662 5232      JMP      RET4     /MESSAGE ROUTINE.
0663 4474      OUT4,    JMS I    XMESS
0664 6145      MSGT4
0665 1060      TAO      CNTALL
0666 3104      DCA      PRNCT
0667 4475      JMS I    XPRINT

//TIME BACKSPACE SHUTDOWN + SETTLE DOWN DELAY//
//
0670 7300      T5,      CLA CLL
0671 1374      TAO      (OUT5      /SETUP T5 EXIT LOCATION.
0672 3071      DCA      OUT        /
0673 4472      RET5,    JMS I    SELALL /GET DRIVE SELECT WORD.
0674 1066      TAO      ODD8000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY.
0675 6705      LCMR     /LOAD COMMAND.
0676 1114      TAO      K7777
0677 6701      LWGR     /LOAD WC = 1.
0678 1115      TAO      K7100
0679 6706      LFGR     /LOAD FUNCTION/ SPACE REVERSE, GO BITS.
0680 6723      SKJD
0683 5302      JMP      .-1      /WAIT FOR JOB DONE.
0684 4437      TIMER    /TIME "BACKSPACE SHUTDOWN + SETTLE DOWN DELAY".
0685 6724      SKTR     /EXIT TIMER ON TUR.
0686 7610      SKP CLA  /USED BY TIMER.
0687 6725      CLF
0690 5273      JMP      RET5     /DO AGAIN ON NEXT DRIVE.
0691 4474      OUT5,    JMS I    XMESS /MESSAGE ROUTINE.
0692 6136      MSGT5
0693 1060      TAO      CNTALL
0694 3104      DCA      PRNCT
0695 4475      JMS I    XPRINT

//TIME READ SHUTDOWN//
//
0716 7300      T6,      CLA CLL
0717 1373      TAO      (OUT6      /SETUP T6 EXIT LOCATION.
0720 3071      DCA      OUT        /
0721 4465      RET6,    JMS I    XWCCA /INITIALIZE WC + CA LOCATIONS.
0722 3777      3777      /CA
0723 7470      -310     /WC
0724 4472      JMS I    SELALL /GET DRIVE SELECT WORD.

```

```

0725 1066 TAD 0D0888 /COMBINE DRIVE SELECT, 888 BPI, ODD PARITY.
0726 6705 LCMR /LOAD COMMAND,
0727 1116 TAD K2100
0730 6706 LFGR /LOAD FUNCTION/ READ, GO BITS,
0731 6723 SKJD
0732 5331 JMP .-1 /WAIT FOR JOB DONE;
0733 6725 CLF /CLEAN FLAGS
0734 4772 JMS I (WRTCNT /WRITE CONTINUOUS
0735 5321 JMP RET6 /DO NEXT DRIVE
0736 4474 OUT6, JMS I XMESS /MESSAGE ROUTINE,
0737 6167 MSGT6
0740 1060 TAD CNTALL
0741 3104 DCA PRNCT
0742 4475 JMS I XPRINT
0743 5744 JMP I LINK7
0744 1000 LINK7, T7

0772 5400
0773 0736
0774 0711
0775 0003
0776 0663
0777 0622
1000 1000

```

PAGE

```

//TIME WRITE XIRG//
//
1000 7300 T7, CLA CLL /SETUP T7 EXIT LOCATION.
1001 1377 TAD (OUT7
1002 3071 DCA OUT
1003 4465 JMS I XWCCA /INITIALIZE WC + CA LOCATIONS,
1004 3776 3776 /CA
1005 7470 -310 /NO
1006 4472 JMS I SELALL /GET DRIVE SELECT WORD;
1007 1066 TAD 0D0888 /COMBINE DRIVE SELECT, 888 BPI, ODD PARITY,
1010 6705 LCMR /LOAD COMMAND,
1011 1117 TAD K4900
1012 6706 LFGR /LOAD FUNCTION/ WRITE XIRG, GO BITS.
1013 4457 TIMER /TIME "WRITE XIRG",
1014 6713 RCAR /EXIT TIMER ON
1015 7710 SPA CLA /CA=4000,
1016 6724 SKTR
1017 5216 JMP .-1 /WAIT FOR T7R,
1020 6725 CLF
1021 5203 JMP RET7 /DO AGAIN ON NEXT DRIVE.
1022 4474 OUT7, JMS I XMESS /MESSAGE ROUTINE
1023 6200 MSGT7
1024 1060 TAD CNTALL
1025 3104 DCA PRNCT
1026 4475 JMS I XPRINT

//TIME LAST CHARACTER TO CU READY//
//
1027 7300 T8, CLA CLL /SETUP T8 EXIT LOCATION.
1030 1376 TAD (OUT8

```

```

1031 3071 RET8, DCA OUT /GET DRIVE SELECT WORD;
1032 4472 JMS I SELALL /COMBINE DRIVE SELECT, 888 BPI, ODD PARITY,
1033 1066 TAD 0D0888 /SAVE THEM,
1034 3106 DCA TEMPI
1035 1106 TAD TEMPI
1036 6705 LCMR /LOAD COMMAND,
1037 1114 TAD K7777 /LOAD WC = -1,
1040 6701 LCMR
1041 1115 TAD K7100 /LOAD FUNCTION/ SPACE REVERSE, GO;
1042 6706 LFGR
1043 6723 SKJD
1044 5243 JMP .-1 /WAIT FOR JOB DONE;
1045 6725 CLF
1046 4465 JMS I XWCCA /INITIALIZE WC & CA LOCATIONS,
1047 3776 3776 /CA
1050 7470 -310 /NO
1051 1106 TAD TEMPI /GET DRIVE, 888 BPI, ODD PARITY,
1052 6705 LCMR /LOAD COMMAND,
1053 1116 TAD K2100
1054 6706 LFGR /LOAD FUNCTION/ READ, GO,
1055 6711 RCAR /READ WC,
1056 7640 SEA CLA /NO = 0?
1057 5255 JMP .-2 /NO, WAIT FOR IT.
1060 4457 TIMER /YES, TIME "LAST CHARACTER TO CU READY,"
1061 6722 SKCB /EXIT TIMER ON CU READY,
1062 7610 SKP CLA
1063 6724 SKTR
1064 5243 JMP .-1 /WAIT FOR T8R,
1065 6725 CLF
1066 5232 JMP RET8 /DO AGAIN ON NEXT DRIVE.
1067 4474 OUT8, JMS I XMESS /MESSAGE ROUTINE
1070 6211 MSGT8
1071 1060 TAD CNTALL
1072 3104 DCA PRNCT
1073 4475 JMS I XPRINT

//TIME READ FROM BOT//
//
1074 7300 T9, CLA CLL /SETUP T9 EXIT LOCATION.
1075 1375 TAD (OUT9
1076 3071 DCA OUT
1077 4463 JMS I XRWND /REWIND ALL SELECTED DRIVES TO BOT,
1078 4465 JMS I XWCCA /INITIALIZE WC & CA LOCATIONS,
1081 3777 3777 /CA
1082 7470 -310 /NO
1083 4472 JMS I SELALL /GET DRIVE SELECT WORD;
1084 1066 TAD 0D0888 /COMBINE DRIVE SELECT, 888 BPI, ODD PARITY,
1085 6705 LCMR /LOAD COMMAND,
1086 1116 TAD K2100
1087 6706 LFGR /LOAD FUNCTION/ READ, GO BITS,
1088 4457 TIMER /TIME "READ FROM BOT,"
1089 6713 RCAR /EXIT TIMER ON CA=4000,
1090 7710 SPA CLA
1091 6724 SKTR
1092 5313 JMP .-1 /WAIT FOR T9R

```

```

1115 6725      CLF
1116 5300      JMP RET9      /DO AGAIN ON NEXT DRIVE
1117 4474      JMS I XMESS /MESSAGE ROUTINE.
1120 6222      MSGT9
1121 1060      TAO CNTALL
1122 3104      DCA PRNCT
1123 4475      JMS I XPRINT

      //TIME SPACE SHUTDOWN + SETTLE DOWN DELAY//
      //
1124 7300      T10, CLA CLL
1125 1374      TAO (OUT10 /SETUP T10 EXIT LOCATION,
1126 3071      DCA OUT
1127 1114      RET10, TAO K7777
1130 6701      LMR /LOAD WC = 1.
1131 4472      JMS I SELALL /GET DRIVE SELECT WORD;
1132 1066      TAO 000000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY,
1133 6705      LMR /LOAD COMMAND
1134 1120      TAO K6100
1135 6706      LFGR /LOAD FUNCTION; SPACE FORWARD, GO BITS,
1136 6723      SKJD
1137 5336      JMP .-1 /WAIT FOR JOB DONE;
1140 4487      TIMER /TIME "SPACE SHUTDOWN + SETTLEDOWN DELAY,"
1141 6724      SKTR /EXIT TIMER ON TUR,
1142 7610      SKP CLA
1143 6725      CLF
1144 5327      JMP RET10 /DO AGAIN ON NEXT DRIVE,
1145 4474      JMS I XMESS /MESSAGE ROUTINE.
1146 6233      MSGT10
1147 1060      TAO CNTALL
1148 3104      DCA PRNCT
1149 4475      JMS I XPRINT
1152 5793      JMP I LINK11
1153 1200      LINK11, T11

1174 1145
1175 1117
1176 1067
1177 1022
1200

```

PAGE

//TIME WRITE EOF//

//

```

1200 7300      T11, CLA CLL
1201 1377      TAO (OUT11 /SETUP T6 EXIT LOCATION,
1202 3071      DCA OUT
1203 4463      JMS I XRWND /REWIND ALL SELECTED DRIVES TO BOT;
1204 4465      JMS I XWCCA /INITIALIZE WC + CA LOCATIONS,
1205 3776      3776 /CA
1206 0000      0000 /WC
1207 4472      JMS I SELALL /GET DRIVE SELECT WORD;
1208 1066      TAO 000000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY,
1211 3106      DCA TEMP1
1212 1106      TAO TEMP1
1213 6705      LMR /LOAD COMMAND,

```

```

1214 1113      TAO K4100
1215 6706      LFGR /LOAD FUNCTION; WRITE, GO BITS,
1216 6723      SKJD
1217 5216      JMP .-1 /WAIT FOR JOB DONE;
1220 6725      CLF
1221 1121      TAO K5100
1222 6706      LFGR /LOAD FUNCTION; WRITE EOF, GO BITS,
1223 4487      TIMER /TIME "WRITE EOF",
1224 6723      SKJD
1225 7610      SKP CLA
1226 6725      CLF
1227 5204      JMP RET11 /DO AGAIN ON NEXT DRIVE;
1230 4474      JMS I XMESS /MESSAGE ROUTINE.
1231 6244      MSGT11
1232 1060      TAO CNTALL
1233 3104      DCA PRNCT
1234 4475      JMS I XPRINT

      //TIME EOF TO EOF SPACE TIME//
      //
1235 7300      T12, CLA CLL
1236 1376      TAO (OUT12 /SETUP T7 EXIT LOCATION
1237 3071      DCA OUT
1240 4463      JMS I XRWND /REWIND ALL SELECTED DRIVES TO BOT;
1241 4465      JMS I XWCCA /INITIALIZE WC + CA LOCATIONS,
1242 3776      3776 /CA
1243 0000      0000 /WC
1244 4472      JMS I SELALL /GET DRIVE SELECT WORD;
1245 1066      TAO 000000 /COMBINE DRIVE SELECT, 800 BPI, ODD PARITY,
1246 3106      DCA TEMP1
1247 1106      TAO TEMP1
1248 6705      LMR /LOAD COMMAND,
1251 1113      TAO K4100
1252 6706      LFGR /LOAD FUNCTION; WRITE, GO BITS,
1253 6723      SKJD
1254 5253      JMP .-1 /WAIT FOR JOB DONE;
1255 6725      CLF
1256 1106      TAO TEMP1
1257 6705      LMR
1260 1122      TAO K1100
1261 6706      LFGR /LOAD FUNCTION, REWIND, GO BITS,
1262 6723      SKJD
1263 5262      JMP .-1
1264 6725      CLF
1265 1106      TAO TEMP1
1266 6705      LMR
1267 1123      TAO K7776
1270 6701      LMR /LOAD WC,
1271 1120      TAO K6100
1272 6706      LFGR /LOAD FUNCTION; SPACE FORWARD, GO BITS;
1273 6711      RMR
1274 1124      TAO K0001 /CHECK FOR WC=1
1275 7640      SEA CLA /INDICATING FIRST RECORD HAS BEEN SPACED OVER,

```

```

1276 5273      JMP      .-3
1277 4487      TIMER
1308 6723      SKJD
1301 7610      SKP CLA
1302 6714      RMSR
1303 0125      AND      K0100
1304 7600      SNA CLA
1305 5317      JMP      ERR12
1306 6725      CLF
1307 5241      JMP      RET12

1310 4474      OUT12, JMS I  XMESS
1311 6255      MSGT12
1312 1060      TAO
1313 3104      DCA
1314 4475      JMS I  XPRINT
1315 5716      JMP I  LINK13
1316 1400      LINK13, T13
1317 1010      ERR12, TAO
1320 1375      TAO      (-2
1321 3010      DCA
1322 3410      DCA I  10
1323 3410      DCA I  10
1324 6725      CLF
1325 5241      JMP      RET12

```

```

/TIME "EOR TO EOF SPACE TIME",
/EXIT TIMER ON JOB DONE,

/CHECK EOF BIT,
/EOF = 1?
/YES, SAVE EOF TIME,
/NO, ZERO TIME COUNTER,

/DO AGAIN ON NEXT DRIVE,

/MESSAGE ROUTINE,

```

```

1375 7776
1376 1310
1377 1230
1400

```

PAGE

```

//TIME WRITE TO ERASE HEAD//
//
1400 7300      T13,   CLA CLL
1401 1377      TAO      {OUT13
1402 3071      DCA      OUT
1403 4463      JMS I  XRWND
1404 4465      RET13, JMS I  XWCCA
1405 3775      3775
1406 7775      -3
1407 4472      JMS I  SELALL
1410 1066      TAO      000000
1411 3106      DCA      TEMP1
1412 1106      TAO      TEMP1
1413 6705      LCMR
1414 1113      TAO      K4100
1415 6706      LFGR
1416 6713      RCAR
1417 1126      TAO      K4001
1418 7640      SEA CLA
1421 5216      JMP      .-3
1422 6712      CLT
1423 1106      TAO      TEMP1

```

```

/SETUP TO EXIT LOCATION,

/REWIND ALL SELECTED DRIVES TO BOT,
/INITIALIZE WC & CA LOCATIONS,
/CA
/NO
/GET DRIVE SELECT WORD,
/COMBINE DRIVE SELECT, 800 BPI, ODD PARITY,
/SAVE IT,

/LOAD COMMAND,

/LOAD FUNCTION, WRITE, GO BITS,
/CA = 3777 INDICATES 2ND WORD SENT,

/2ND WORD SENT?
/NO, GO AGAIN,
/YES, ISSUE CLEAR ALL,

```

```

1424 6705      LCMR
1425 1122      TAO      K1100
1426 6706      LFGR
1427 6714      RMSR
1428 0127      AND      K1000
1431 7600      SNA CLA
1432 5227      JMP
1433 6725      CLF      .-5
1434 4465      JMS I  XWCCA
1435 3776      3776
1436 7775      -3
1437 1106      TAO      TEMP1
1438 6705      LCMR
1439 1116      TAO      K2100
1442 6706      LFGR
1443 6713      RCAR
1444 1126      TAO      K4001
1445 7640      SEA CLA
1446 5243      JMP      .-3
1447 4487      TIMER
1448 6713      RCAR
1449 7710      SPA CLA
1452 6723      SKJD
1453 5202      JMP      .-1
1454 6725      CLF
1455 5204      JMP      RET13
1456 4474      OUT13, JMS I  XMESS
1457 6266      MSGT13
1460 1060      TAO
1461 3104      DCA
1462 4475      JMS I  XPRINT

```

```

/LOAD COMMAND

/LOAD FUNCTION, REWIND, GO BITS,

/CHECK FOR BOT,

/INITIALIZE WC & CA LOCATIONS
/CA
/NO
/LOAD COMMAND,

/LOAD FUNCTION, READ, GO BITS,
/CA=3777 INDICATES 2ND WORD READ

/2ND WORD READ?
/NO, WAIT,
/YES, TIME "WRITE TO ERASE HEAD,"
/EXIT TIMER ON CA=4000,

```

```

//ONE INCH DATA TIME//
//
1463 7300      T14,   CLA CLL
1464 1376      TAO      {OUT14
1465 3071      DCA      OUT
1466 4463      JMS I  XRWND
1467 4465      RET14, JMS I  XWCCA
1470 3776      3776
1471 7160      -820
1472 4472      JMS I  SELALL
1473 1066      TAO      000000
1474 6705      LCMR
1475 1113      TAO      K4100
1476 6706      LFGR
1477 6713      RCAR
1500 7700      SNA CLA
1501 5277      JMP      .-2
1502 4487      TIMER
1503 6711      RWQR
1504 7600      SNA CLA
1505 6724      SKTR
1506 5305      JMP      .-1

```

```

/DO AGAIN ON NEXT DRIVE,
/MESSAGE ROUTINE,

/SETUP TO EXIT LOCATION,

/REWIND ALL SELECTED DRIVES TO BOT,
/INITIALIZE WC & CA LOCATIONS,
/CA
/NO
/GET DRIVE SELECT WORD,
/COMBINE DRIVE SELECT, 800 BPI, ODD PARITY
/LOAD COMMAND,

/LOAD FUNCTION, WRITE, GO BITS,
/MINUS CA INDICATES FIRST WORD TAKEN,

/TIME "800 BPI DATETIME",
/EXIT TIMER ON WC=0,

/WAIT FOR PUR,

```

```

1507 6725 CLF
1510 5267 JMP RET14
1511 4474 OUT14, JMS I XMESS
1512 6277 MSGT14
1513 1000 TAO CNTALL
1514 3104 DCA PRNCT
1515 4475 JMS I XPRINT
1516 5717 JMP I LINKGP
1517 1000 LINKGP, GAPS

```

```

/DO AGAIN ON NEXT DRIVE,
/MESSAGE ROUTINE,

```

```

1576 1511
1577 1486
1578 1000

```

PAGE

//CREATE GAPS TO BE TIMED IN GAP CONSISTENCY TEST//

//

```

1600 7300 GAPS, CLA CLL
1601 1377 TAO (OUTGAP
1602 3071 DCA OUT
1603 4463 JMS I XRWND
1604 1133 RETGAP, TAO KM2
1605 3306 DCA GAP2CT
1606 1133 TAO KM2
1607 3307 DCA GAP5CT
1608 1134 TAO KM3
1609 3310 DCA GAP6CT
1610 1102 TAO KM4
1611 3311 DCA GAP7CT
1612 1313 TAO KM5
1613 3312 DCA GAP8CT
1614 4465 JMS I XNCCA
1615 3767 3767
1616 7470 -310
1617 4472 JMS I SELALL
1618 1066 TAO ODD000
1619 3106 DCA TEMP1
1620 1106 TAO TEMP1
1621 6705 LCMR
1622 1113 TAO K4100
1623 6706 LFGR
1624 6723 SKJD
1625 5230 JMP
1626 6725 DOGAP2, CLF
1627 4465 JMS I XNCCA
1628 3767 3767
1629 7470 -310
1630 1106 TAO TEMP1
1631 6705 LCMR
1632 1113 TAO K4100
1633 6706 LFGR
1634 6723 SKJD
1635 5242 JMP
1636 6724 SKTR
1637 5244 JMP

```

```

/SETUP GAPS EXIT LOCATION,
/REWIND ALL SELECTED DRIVES TO BOT,
/GAP2 COUNT=-2,
/GAP5 COUNT=-2,
/GAP6 COUNT=-3,
/GAP7 COUNT=-4,
/GAP8 COUNT=-5,
/INITIALIZE WC, CA LOCATIONS,
/CA
/WC
/GET DRIVE SELECT WORD,
/COMBINE DRIVE SELECT, 000 BPI, 000 PARITY,
/SAVE IT,
/LOAD COMMAND,
/LOAD FUNCTION/ WRITE, 60 BITS,
/WAIT FOR JOB DONE,
/WRITE NONSTOP, GAP1 CREATED,
/INITIALIZE WC, CA LOCATIONS,
/CA
/WC
/PICKUP DRIVE, BPI, ODD PARITY,
/LOAD COMMAND,
/LOAD FUNCTION/ WRITE 60 BITS,
/WAIT FOR JOB DONE,
/WAIT FOR PUR,

```

```

1646 2306 ISZ GAP2CT
1647 5232 JMP DOGAP2
1648 6725 CLF
1649 4936 JMS I XBAK
1650 4940 JMS I XREAD
1651 4937 JMS I XRITE1
1652 4937 JMS I XRITE1
1653 4936 JMS I XBAK
1654 4937 JMS I XRITE1
1655 4936 JMS I XBAK
1656 4937 JMS I XRITE1
1657 4937 JMS I XRITE1
1658 4936 DOGAP5, JMS I XBAK
1659 4937 JMS I XRITE1
1660 2307 ISZ GAP5CT
1661 5260 JMP DOGAP5
1662 4937 JMS I XRITE1
1663 4936 DOGAP6, JMS I XBAK
1664 4937 JMS I XRITE1
1665 2310 ISZ GAP6CT
1666 5265 JMP DOGAP6
1667 4937 JMS I XRITE1
1668 4936 DOGAP7, JMS I XBAK
1669 4937 JMS I XRITE1
1670 2311 ISZ GAP7CT
1671 5272 JMP DOGAP7
1672 4937 JMS I XRITE1
1673 4936 DOGAP8, JMS I XBAK
1674 4937 JMS I XRITE1
1675 2312 ISZ GAP8CT
1676 5277 JMP DOGAP8
1677 5204 JMP RETGAP
1678 5705 OUTGAP, JMP I LINK15
1679 2000 LINK15, T15
1680 0000 GAP2CT, 0
1681 0000 GAP5CT, 0
1682 0000 GAP6CT, 0
1683 0000 GAP7CT, 0
1684 0000 GAP8CT, 0
1685 7773 XMS, -5

```

```

/WRITE 3RD RECORD START-STOP
/TO CREATE GAP2,
/SPACE REVERSE 1 RECORD,
/READ RECORD,
/WRITE 4TH RECORD/ 3RD GAP CREATED,
/START GAP4, WRITE 1 RECORD,
/SPACE REVERSE OVER RECORD JUST WRITTEN,
/WRITE 5TH RECORD/ 4TH GAP CREATED,
/START GAP5, WRITE 1 RECORD,
/SPACE REVERSE OVER RECORD JUST WRITTEN,
/WRITE 1 RECORD,
/DO XBAK + XRITE 2 MORE TIMES
/TO CREATE 6TH GAP,
/START GAP6, WRITE 1 RECORD,
/SPACE REVERSE OVER RECORD JUST WRITTEN,
/WRITE 1 RECORD,
/DO XBAK + XRITE 3 MORE TIMES
/TO CREATE 7TH GAP,
/START GAP7, WRITE 1 RECORD,
/SPACE REVERSE OVER RECORD JUST WRITTEN,
/WRITE 1 RECORD,
/DO XBAK + XRITE 4 MORE TIMES
/TO CREATE 8TH GAP,
/DO AGAIN ON NEXT DRIVE,

```

```

1777 1704
1778 2000

```

PAGE

```

//NOW READ NONSTOP THE GAPS JUST CREATED
//TYPE GAP TIME AS EACH GAP IS READ ON ALL DRIVES
//GAP1 > GAP2, GAP2 APPROXIMATELY 8 GAPS
//GAP4 THRU GAPS SHOULD GET INCREASINGLY LONGER
//

```

```

1800 7300 T15, CLA CLL
1801 1377 TAO (OUT15A
1802 3071 DCA OUT
1803 1264 TAO M7
1804 3263 DCA GAPCNT

```

/SETUP T15A EXIT LOCATION,

```

2005 1376      TAQ      (GAPTRL=1
2006 3812      DCA      INDX12
2007 4463      JMS I    XRWND
2008 4472      RET15A,  JMS I    SELALL
2009 1066      TAD      ODB800
2010 3106      DCA      TEMP1
2011 4946      JMS I    XREAD2
2012 4941      JMS I    XREAD1
2013 5210      JMP      RET15A
2014 4250      OUT15A,  JMS      GAPOUT
2015 1375      TAQ      (OUT15B
2016 3871      DCA      OUT
2017 4472      RET15B,  JMS I    SELALL
2018 1066      TAD      ODB800
2019 3106      DCA      TEMP1
2020 4936      JMS I    XBAK
2021 4946      JMS I    XREAD2
2022 4941      JMS I    XREAD1
2023 5221      JMP      RET15B
2024 4250      OUT15B,  JMS      GAPOUT
2025 2263      JMS      GAPCNT
2026 5221      JMP      RET15B
2027 1061      TAQ      CNT7
2028 7600      SNA CLA
2029 5945      JMP I    XEND
2030 5637      JMP I    LINK56
2031 2200      LINK56,  SET556
2032 0261      GAPTRL,  0261
2033 0262      0262
2034 0263      0263
2035 0264      0264
2036 0265      0265
2037 0266      0266
2038 0267      0267
2039 0270      0270
2040 0000      GAPOUT,  0
2041 4474      JMS I    XMESS
2042 6310      MSGT15
2043 1412      TAD I    INDX12
2044 4901      JMS I    XOTY
2045 4474      JMS I    XMESS
2046 6313      MSGSPC
2047 1060      TAQ      CNTALL
2048 3104      DCA      PRNCT
2049 4475      JMS I    XPRINT
2050 5650      JMP I    GAPOUT
2051 0000      GAPCNT,  0
2052 7771      M7,     -7
2175 2030
2176 2037
2177 2016
2200

```

PAGE

```

2200 7300      SET556,  CLA CLL
2201 4474      JMS I    XMESS
2202 6321      MSG556
2203 4456      JMS I    XHEAD1
2204 4451      JMS I    XCRLF
2205 1142      TAQ      K0400
2206 7001      IAC
2207 3143      DCA      XBPI
2210 1213      TAQ      M426
2211 3147      DCA      INCH
2212 5544      JMP I    LINK16
2213 7352      M426,   -426
2214 7300      //TIME WRITE FROM BOT DELAY AT EITHER 556 OR 200 BPI//
2215 1377      //
2216 3871      T16,    CLA CLL
2217 4464      TAQ      (OUT16
2218 4465      DCA      OUT
2219 3776      RET16,  JMS I    XRWND7
2220 7470      JMS I    XWCCA
2221 4473      JMS I    SELT7
2222 1143      TAD      XBPI
2223 6705      LCMR
2224 1113      TAD      K4100
2225 6706      LFGP
2226 4457      TIMER
2227 6713      RCAR
2228 7710      SPA CLA
2229 6724      SKTR
2230 5233      JMP      =I
2231 6725      CLP
2232 5220      JMP      RET16
2233 4474      OUT16,  JMS I    XMESS
2234 6112      MSGT1
2235 1061      TAQ      CNT7
2236 3104      DCA      PRNCT
2237 4475      JMS I    XPRINT
2244 7300      T17,    CLA CLL
2245 1376      TAQ      (OUT17
2246 3871      DCA      OUT
2247 4464      JMS I    XRWND7
2248 4465      JMS I    XWCCA
2249 3776      JMS I    SELT7
2250 7470      JMS I    SELT7
2251 4473      JMS I    SELT7

```

```

0047 1111      TAQ      NCBU      /SUBTRACT MIKECT FROM ORIGINAL CONSTANT,
0050 7040      CMA
0051 1112      TAQ      PCPU      /NOW ADD ORIGINAL CONSTANT TO NUMBER OF
0052 3256      DCA      DIVDEND   /ITERATIONS IN MIKECT TO FORM DIVDEND,
0053 1112      TAQ      PCPU      /NOW SETUP
0054 3257      DCA      DIVSR     /DIVISOR,
0055 4274      JMS      DIVIDE    /ENTER DIVIDE ROUTINE,

0056 0000      DIVDEND, 0
0057 0000      DIVSR, 0
0060 7200      DIVRET, CLA
0061 1103      TAQ      HDIVND    /DIVDEND,
0062 3273      DCA      SAVANS    /DIVISOR,
0063 1110      TAQ      MILICT    /RETURN HERE FROM DIVIDE ROUTINE,
0064 3410      DCA I   INDX10    /GET ANSWER,
0065 1273      TAQ      SAVANS    /SAVE IT,
0066 3410      DCA I   INDX10    /GET VALUE IN MILLISECONDS,
0067 3107      DCA      MIKECT   /STORE AHAP,
0070 3110      DCA      MILICT   /GET VALUE IN MICRO SECONDS,
0071 3505      DCA I   NULL      /STORE AHAP,
0072 5611      JMP I   TIME      /CLEAN TIME COUNTERS,
0073 0000      SAVANS, 0        /RETURN TO TEST,

/SINGLE PRECISION DIVIDE SUBROUTINE

```

```

0074 0000      DIVIDE, 0
0075 7100      CLL
0076 3103      DCA      HDIVND    /HIGH ORDER DIVDEND,
0077 1674      TAQ I   DIVIDE    /FETCH LOW ORDER DIVDEND,
0080 3326      DCA      LDIVND    /SAVE IT,
0081 2274      ISR      DIVIDE    /UPDATE POINTER,
0082 1674      TAQ I   DIVIDE    /FETCH DIVISOR,
0083 3327      DCA      DIVSR     /SAVE IT,
0084 2274      ISR      DIVIDE
0085 1331      TAQ      KM13
0086 3330      DCA      DIVCNT
0087 5320      JMP      DV2
0088 1103      DV3, TAQ      HDIVND /DIVIDE LOOP
0089 7004      RAL
0090 3103      DCA      HDIVND    /DIVDEND LEFT SHIFT,
0091 1103      TAQ      HDIVND
0092 1327      TAQ      DIVSR     /COMPARE DIVISOR/DIVDEND,
0093 7430      SEL
0094 3103      DCA      HDIVND    /REMAINDER AFTER SUBTRACT,
0095 7200      CLA
0096 1326      DV2, TAQ      LDIVND /QUOTIENT BITS,
0097 7004      RAL              /ENTER HERE,
0098 3326      DCA      LDIVND
0099 2330      ISR      DIVCNT    /DONE 127
0100 5310      JMP      DV3      /NO, CONTINUE,
0101 5674      JMP I   DIVIDE    /YES, EXIT,

0126 0000      LDIVND, 0

```

```

0127 0000      DIVSR, 0
0130 0000      DIVCNT, 0
0131 7765      KM13, -13
0132 0000      XNULL, 0

0175 0142
0176 7636
0177 7637
0180 5200      PAGE

0200 0000      RWIND, 0 /ROUTINE TO REWIND ALL SELECTED TRANSPORTS TO BOT/
0201 6722      SKGB
0202 5201      JMP      .-1      /WAIT FOR BUR,
0203 6725      CLF
0204 1377      TAQ      (ALLTU   /SETUP DRIVE SELECTION
0205 3070      DCA      DRVSEL    /POINTER FOR ALL DRIVES,
0206 1060      TAQ      CNTALL    /SETUP COUNT
0207 7041      CIA
0208 3261      DCA      REWCNT    /TO INDICATE WHEN ALL
0209 1257      TAQ      EXALL     /DRIVES HAVE BEEN REWOUND,
0210 3256      DCA      GETBAK
0211 1066      TAQ      OD0000    /PICKUP PARITY AND SPI,
0212 3263      DCA      TEMPS    /SAVE THEM,
0213 5233      JMP      RWINDA    /GO AND REWIND SELECTED DRIVES,
0214 0000      RWIND7, 0
0215 6722      SKGB
0216 5217      JMP      .-1      /WAIT FOR BUR,
0217 6725      CLF
0218 1376      TAQ      (TU7     /SETUP DRIVE SELECTION
0219 3070      DCA      DRVSEL    /POINTER FOR 7 TRACK DRIVES ONLY,
0220 1061      TAQ      CNT7     /SETUP COUNT TO INDICATE
0221 7041      CIA
0222 3261      DCA      REWCNT    /WHEN ALL 7 TRACK DRIVES
0223 1260      TAQ      EXTU7    /HAVE BEEN REWOUND,
0224 3256      DCA      GETBAK
0225 1143      TAQ      XBP1     /PICKUP PARITY AND SPI,
0226 3263      DCA      TEMPS    /SAVE THEM,
0227 1470      RWINDA, TAQ I   DRVSEL /GET DRIVE SELECTED,
0228 1263      TAQ      TEMPS    /COMBINE WITH PARITY AND SPI,
0229 6705      LCMR
0230 6724      SKTR
0231 5236      JMP      .-1      /LOAD COMMAND,
0232 6714      RMR
0233 0262      AND      MK1000
0234 7640      SEA CLA
0235 5252      JMP      .+7
0236 1375      TAQ      (1100
0237 6706      LFOR
0238 6723      SKJD
0239 5246      JMP      .-1      /LOAD FUNCTION/ REWIND, GO BITS,
0240 6724      SKTR
0241 5250      JMP      .-1      /WAIT FOR JOB DONE,
0242 6725      CLF
0243 5250      JMP      .-1      /WAIT FOR BUR,
0244 6725      CLF

```

```

2253 2070      ISR      DRVSEL
2254 2261      ISR      RENCNT      /ALL DRIVES DONE?
2255 2233      JMP      RHINDA      /NO, CONTINUE,
2256 7402      GETBAK, HLT      /MODIFIED TO JMP1,

2257 5600      EXALL, JMP I  RWIND
2260 5616      EXTU7, JMP I  RWIND?
2261 0000      RENCNT, 0
2262 1000      MK1000, 1000
2263 0000      TEMPS, 0

      /ROUTINE USED TO SELECT BOTH 7 AND 9 TRACK DRIVES/
      /TO BE TESTED AT 800 BPI/
      /
2264 0000      ALTU79, 0
2265 2050      ISR      ONCE      /DO INITIAL SETUP ONE TIME ONLY,
2266 5274      JMP      ,+6
2267 1377      TAO      (ALLTU      /INITIALIZE SELECT TABLE,
2270 3070      DCA      DRVSEL
2271 1060      TAO      CNTALL
2272 7040      CMA
2273 3324      DCA      SELCNT
2274 1470      TAO I    DRVSEL      /GET DRIVE TO BE TESTED,
2275 2070      ISR      DRVSEL      /UPDATE POINTER,
2276 2324      ISR      SELCNT      /HAVE ALL DRIVES BEEN TESTED?
2277 5664      JMP I    ALTU79      /NO, CONTINUE,
2280 7200      CLA      /YES, EXIT CURRENT TEST,
2281 1114      TAO      K7777      /SETUP INITIALIZATION
2282 3050      DCA      ONCE      /LOCATION,
2283 5471      JMP I    OUT      /EXIT.

      /ROUTINE USED TO SELECT 7 TRACK DRIVES TO BE/
      /TESTED AT 556 AND 200 BPI/
      /
2304 0000      ALLTU7, 0
2305 2050      ISR      ONCE      /DO INITIAL SETUP ONE TIME ONLY,
2306 5314      JMP      ,+6
2307 1376      TAO      (TU7      /INITIALIZE
2310 3070      DCA      DRVSEL
2311 1061      TAO      CNT7
2312 7040      CMA
2313 3324      DCA      SELCNT
2314 1470      TAO I    DRVSEL      /GET DRIVE TO BE TESTED,
2315 2070      ISR      DRVSEL      /UPDATE POINTER,
2316 2324      ISR      SELCNT      /HAVE ALL DRIVES BEEN TESTED?
2317 5704      JMP I    ALLTU7      /NO, CONTINUE,
2320 7200      CLA      /YES, EXIT CURRENT TEST,
2321 1114      TAO      K7777      /SETUP INITIALIZATION
2322 3050      DCA      ONCE      /LOCATION,
2323 5471      JMP I    OUT      /EXIT.

2324 0000      SELCNT, 0

```

```

      /ROUTINE USED TO OUTPUT TO TTY BOTH 7 AND 9 TRACK/
      /DRIVES TESTED AT 800 BPI/
      /
2325 0000      ALLMSG, 0
2326 1374      TAO      (ALLTBL      /INITIALIZE SELECTION TABLE,
2327 3070      DCA      DRVSEL
2330 1060      TAO      CNTALL      /SETUP COUNTER/TOTAL NUMBER OF DRIVES
2331 7041      CIA      /SELECTED,
2332 3324      DCA      SELCNT
2333 4474      JMS I    XMESS      /MESSAGE SUBROUTINE,
2334 6076      MSG06      /UNIT,
2335 1470      TAO I    DRVSEL      /GET UNIT NUMBER,
2336 4501      JMS I    XQTY      /OUTPUT TO TTY,
2337 1365      TAO      SPACE
2340 4501      JMS I    XQTY
2341 2070      ISR      DRVSEL      /UPDATE POINTER,
2342 2324      ISR      SELCNT      /ARE ALL UNITS OUTPUT?
2343 5333      JMP      ALLRET      /NO, GET NEXT UNIT,
2344 5725      JMP I    ALLMSG      /YES, EXIT.

      /ROUTINE USED TO OUTPUT TO TTY ALL 7 TRACK DRIVES/
      /TESTED AT 556 AND 200 BPI/
      /
2345 0000      TU7MSG, 0
2346 1373      TAO      (TU7TBL      /INITIALIZE SELECTION TABLE,
2347 3070      DCA      DRVSEL
2350 1061      TAO      CNT7      /SETUP COUNTER/NUMBER OF 7 TRACK
2351 7041      CIA      /DRIVES SELECTED,
2352 3324      DCA      SELCNT
2353 4474      JMS I    XMESS      /MESSAGE SUBROUTINE,
2354 6076      MSG06      /UNIT,
2355 1470      TAO I    DRVSEL      /GET UNIT NUMBER,
2356 4501      JMS I    XQTY      /OUTPUT TO TTY,
2357 1365      TAO      SPACE
2360 4501      JMS I    XQTY
2361 2070      ISR      DRVSEL      /UPDATE POINTER,
2362 2324      ISR      SELCNT      /ARE ALL UNITS OUTPUT?
2363 5333      JMP      TU7RET      /NO, GET NEXT UNIT,
2364 5745      JMP I    TU7MSG      /YES, EXIT.

2365 0240      SPACE, 0240

2373 0170
2374 0160
2375 1100
2376 0040
2377 0030
2400 5400      PAGE

```

```

//ROUTINE TO WRITE CONTINUOUS AND TIME TILL LEADING EDGE OF DOWN.//
//THIS ROUTINE USED FOR TIMING WRITE AND READ SHUTDOWN.//
2400 0000      WRCNT, 0

```

```

2401 1377      TAO      (3      /ADJUST FOR SETUP TIME
2402 3505      DCA I     NULL
2403 1113      TAO      K4100   /CONTINUOUS WRITE
2404 6706      LFER
2405 4457      TIMER
2406 6711      RWCR
2407 7640      SEA CLA
2410 6723      SKJD
2411 5210      JMP      ,=1
2412 6724      SKTR
2413 5212      JMP      ,=1
2414 6725      CLF
2415 5600      JMP I     WRTCNT   /CLEAN ALL
                                           /EXIT

```

/SINGLE CHARACTER OUTPUT ROUTINE
/ENTERED WITH CHARACTER IN AC

```

2416 0000      QTY,      0
2417 6046      TLF
2420 6041      TSF
2421 5220      JMP      ,=1
2422 6042      TCF
2423 7200      CLA
2424 5616      JMP I     QTY

```

```

2425 0000      PRINT,    0
2426 7300      CLA CLL
2427 1376      TAO      (6377
2430 3010      DCA      INDX10
2431 1104      TAO      PRNCT
2432 7041      CIA
2433 3104      DCA      PRNCT
2434 3077      DCA      VFS20
2435 3100      DCA      VFSUPR
2436 1410      TAO I     INDX10
2437 4476      JMS I     XDEC
2440 1277      TAO      K256
2441 4501      JMS I     XOTY
2442 1300      TAO      K7775
2443 3077      DCA      VFS20
2444 7040      CMA
2445 3100      DCA      VFSUPR
2446 1410      TAO I     INDX10
2447 4476      JMS I     XDEC
2450 1276      TAO      K240
2451 4501      JMS I     XOTY
2452 2104      ISZ      PRNCT
2453 5234      JMP      LOPRIN
2454 4260      JMS      CRLF
2455 1376      TAO      (6377
2456 3010      DCA      INDX10
2457 5625      JMP I     PRINT

```

/ROUTINE TO OUTPUT ACCUMULATED
/TIMES FOR ALL DRIVES,

/SETUP PRINT COUNTER,

/DECIMAL POINT,

/SPACE,

```

2460 0000      CRLF,      0
2461 1131      TAO      K215
2462 6046      TLF
2463 6041      TSF
2464 5263      JMP      ,=1
2465 6042      TCF
2466 7200      CLA
2467 1130      TAO      K212
2470 6046      TLF
2471 6041      TSF
2472 5271      JMP      ,=1
2473 6042      TCF
2474 7200      CLA
2475 5660      JMP I     CRLF

```

/CARRIAGE RETURN,

/LINE FEED,

```

2476 0240      K240,      0240
2477 0256      K256,      0256
2500 7775      K7775,      7775

```

/UNSIGNED DECIMAL PRINT
/CALL WITH NUMBER TO BE TYPED IN AC
/RETURN TO LOCATION FOLLOWING THE JMS.

```

2501 0000      DECPRT,    0
2502 3363      DCA      VALUE
2503 3364      DCA      DIGIT
2504 3366      DCA      VTEMP2
2505 1102      TAO      KM4
2506 3365      DCA      VTEMP1
2507 1356      TAO      ADDR2A
2510 3315      DCA      ARROW
2511 7410      SKP
2512 3363      DCA      VALUE
2513 7100      CLL
2514 1363      TAO      VALUE
2515 1357      TAO      TENPWR
2516 7430      SEL
2517 2364      ISZ
2520 7430      SEL
2521 5312      JMP      ARROW=3
2522 7200      CLA
2523 1077      TAO      VFS20
2524 7650      SNA CLA
2525 5330      JMP      ,+3
2526 2077      ISZ      VFS20
2527 5347      JMP      LOOPD+2
2530 1364      TAO      LOOPD
2531 7450      SNA
2532 5335      JMP      ,+3
2533 2366      ISZ      VTEMP2
2534 5345      JMP      LOOPD
2535 1366      TAO      VTEMP2
2536 7640      SEA CLA
2537 5345      JMP      LOOPD
2540 1100      TAO      VFSUPR

```

/SAVE INPUT,

/CLEAN,

/SET COUNTER TO 4,

/SET TABLE POINTER,

/SAVE

/SUBTRACT POWER OF TEN,

/DEVELOP BCD DIGIT,

/LOOP

/HAVE BCD DIGIT,

/SUPPRESS LEADING DIGITS?

/NO

/YES, COUNT AND DO NOT PRINT,

/GET DIGIT

/IS THIS A ZERO?

/YES

/NO

/HAS A DIGIT BEEN TYPED?

/YES,

/NO,

```

3541 7648 SEA CLA /SHOULD THIS ONE BE SUPPRESSED?
3542 9345 JMP LOOPD /NO.
3543 1276 TAO K240 /YES, TYPE SPACE
3544 9346 JMP LOOPD*1
3545 1367 LOOPD, TAO K260 /MAKE IT ASCII
3546 4901 JMS I XDTY /TYPE DIGIT.
3547 3364 DCA DIGIT /CLEAN
3550 2315 ISE ARROW /UPDATE POINTER.
3551 2365 ISE VTEMP1 /DONE ALL FOUR?
3552 5314 JMP ARROW-1 /NO, CONTINUE.
3553 3100 DCA VFSUPR
3554 3077 DCA VFS20
3555 9701 JMP I DECPRT /YES, EXIT.

3556 1357 ADDR2A, TAO TENPWR
3557 6030 TENPWR, =1750 /ONE THOUSAND
3558 7634 =0144 /ONE HUNDRED
3559 7766 =0012 /TEN
3560 7777 =0001 /ONE

3563 0000 VALUE, 0
3564 0000 DIGIT, 0
3565 0000 VTEMP1, 0
3566 0000 VTEMP2, 0
3567 0260 K260, 260

3576 6377
3577 0003
3580 5600

```

PAGE

//ROUTINE TO WRITE ONE 200(10) WORD RECORD//

```

//
3600 0000 RITE1, 0
3601 4465 JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
3602 3777 /CA
3603 7470 =310 /WC
3604 1106 TAO TEMP1 /PICKUP DRIVE, BPI, ODD PARITY.
3605 6705 LCHR /LOAD COMMAND.
3606 1113 TAO K4100
3607 6706 LFGR /LOAD FUNCTION/ WRITE, GO BITS,
3610 6723 SKJD
3611 5210 JMP ,=1 /WAIT FOR JOB DONE.
3612 6724 SKTR
3613 5212 JMP ,=1 /WAIT FOR TUR.
3614 6725 CLP
3615 5600 JMP I RITE1 /RETURN.

```

//BACKSPACE ONE RECORD//

```

//
3616 0000 BAKSPA, 0
3617 1132 TAO KM1
3620 6701 LCHR /WC=1
3621 1106 TAO TEMP1 /SET DRIVE, BPI, ODD PARITY.
3622 6705 LCHR /LOAD COMMAND.
3623 1113 TAO K7100

```

```

3624 6706 LFGR /LOAD FUNCTION/ SPACE REVERSE, GO.
3625 6723 SKJD
3626 5225 JMP ,=1 /WAIT FOR JOB DONE.
3627 6724 SKTR
3630 5227 JMP ,=1 /WAIT FOR JOB DONE.
3631 6725 CLP
3632 5616 JMP I BAKSPA /RETURN.

```

//ROUTINE TO READ 1 RECORD START-STOP//

```

//
3633 0000 READ, 0
3634 4465 JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
3635 3777 /CA
3636 7470 =310 /WC
3637 1106 TAO TEMP1 /PICKUP DRIVE, BPI, ODD PARITY.
3640 6705 LCHR /LOAD COMMAND.
3641 1116 TAO K2100
3642 6706 LFGR /LOAD FUNCTION/ READ, GO BITS,
3643 6723 SKJD
3644 5243 JMP ,=1 /WAIT FOR JOB DONE.
3645 6724 SKTR
3646 5245 JMP ,=1 /WAIT FOR TUR.
3647 6725 CLP
3650 5633 JMP I READ /RETURN.

```

//ROUTINE TO READ 1 RECORD AND TIME THE INTERRECORD
//GAP FOLLOWING IT

```

//
3651 0000 READ1, 0
3652 4465 JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
3653 3777 /CA
3654 7470 =310 /WC
3655 1106 TAO TEMP1 /PICKUP DRIVE, BPI, ODD PARITY.
3656 6705 LCHR /LOAD COMMAND.
3657 1116 TAO K2100
3660 6706 LFGR /LOAD FUNCTION/ READ, GO BITS.
3661 4497 TIMER /TIME GAP.
3662 6713 RGRAR /EXIT TIMER ON C1=4000.
3663 7710 SPA CLA
3664 6724 SKTR
3665 5264 JMP ,=1 /RETURN HERE.
3666 6725 CLP /WAIT FOR TUR.
3667 5691 JMP I READ1 /RETURN.

```

//ROUTINE TO READ 1 RECORD NONSTOP//

```

//
3670 0000 READ2, 0
3671 4465 JMS I XWCCA /INITIALIZE WC, CA LOCATIONS.
3672 3777 /CA
3673 7470 =310 /WC
3674 1106 TAO TEMP1 /PICKUP DRIVE, BPI, ODD PARITY.
3675 6705 LCHR /LOAD COMMAND.
3676 1116 TAO K2100
3677 6706 LFGR /LOAD FUNCTION/ READ, GO BITS,
3680 6723 SKJD

```

```

9701 5300      JMP      ,+1      /WAIT FOR JOB DONE,
9702 6725      CLF
9703 5670      JMP I   READ2     /RETURN.

```

```

//ENDING ROUTINE//
9704 7300      FINI.   CLA CLL
9705 4474      JMS I   XMESS
9706 6343      MSGEND
9707 5710      JMP I   ,+1
9710 0206      SELECT

```

```

/MESSAGE TYPE-OUT ROUTINE
/

```

```

9711 0000      MESSAGE, 0
9712 7240      CLA CMA
9713 1711      TAD I   MESSAGE
9714 3011      DCA     11
9715 2311      ISZ    MESSAGE
9716 1411      TAD I   11
9717 3330      DCA     MSGRGHT
9720 1330      TAD     MSGRGHT
9721 7012      RTR
9722 7012      RTR
9723 7012      RTR
9724 4331      JMS     TYPECH
9725 1330      TAD     MSGRGHT
9726 4331      JMS     TYPECH
9727 5316      JMP     MESSAGE+5
9730 0000      MSGRGHT, 0
9731 0000      TYPECH, 0
9732 0363      ANQ     MASK77
9733 7450      SNA
9734 5711      JMP I   MESSAGE
9735 1364      TAD     M40
9736 7510      SPA
9737 5342      JMP     ,+3
9740 1365      TAD     C240
9741 5355      JMP     MTP
9742 7001      IAC
9743 7440      SEA
9744 5347      JMP     ,+3
9745 1366      TAD     C215
9746 5355      JMP     MTP
9747 7001      IAC
9748 7440      SEA
9749 5354      JMP     ,+3
9752 1367      TAD     C212
9753 5355      JMP     MTP
9754 1370      TAD     C336
9755 6046      MTP,   TL9
9756 6041      TSF
9757 5356      JMP     ,+1
9760 6042      TCF

```

```

9761 7200      CLA
9762 5731      JMP I   TYPECH
9763 0077      MASK77, 77
9764 7740      M40,   =40
9765 0240      C240,  240
9766 0215      C215,  215
9767 0212      C212,  212
9770 0336      C336,  336

```

```

6000      PAGE

```

```

0000 3736      MSG1,   TEXT    "••IMBE DRIVE FUNCTION TIMER••"
0001 2415
0002 7005
0003 4004
0004 2211
0005 2005
0006 4006
0007 2516
0010 0324
0011 1117
0012 1040
0013 2411
0014 1905
0015 2237
0016 3600

```

```

0017 3736      MSG2,   TEXT    "••MAINDEC-08-DHMC-A••"
0020 1901
0021 1116
0022 0405
0023 0355
0024 6070
0025 5504
0026 1024
0027 1903
0030 5501
0031 3736
0032 0000

```

```

0033 3736      MSG3,   TEXT    "••SELECT DRIVES••"
0034 2305
0035 1405
0036 0324
0037 4004
0040 2211
0041 2005
0042 2337
0043 3600

```

```

0044 3736      MSG4,   TEXT    "••SELECT 7 AND/OR 9 TRACK OPERATION••"
0045 2305
0046 1405
0047 0324

```

```

0050 4067
0051 4001
0052 1604
0053 5717
0054 2240
0055 7140
0056 2422
0057 0103
0060 1348
0061 1720
0062 0922
0063 0124
0064 1117
0065 1637
0066 3600

0067 3736 MSG5, TEXT "FUNCTION="
0070 0625
0071 1603
0072 2411
0073 1716
0074 3736
0075 0000

0076 4040 MSG6, TEXT " UNIT"
0077 2516
0100 1124
0101 0000

0102 7060 MSG7, TEXT "800 BPI "
0103 6040
0104 0220
0105 1140
0106 4040
0107 4040
0110 4040
0111 4000

0112 2722 MSG11, TEXT "MR FM BOT DELAY "
0113 4006
0114 1540
0115 0217
0116 2440
0117 0405
0120 1401
0121 3140
0122 0000

0123 2722 MSG12, TEXT "WRITE SHUTDOWN "
0124 1124
0125 0540
0126 2310
0127 2524
0130 0417
0131 2716

```

```

0132 4040
0133 0000

0134 2722 MSG13, TEXT "WRITE START "
0135 1124
0136 0540
0137 2324
0140 0122
0141 2440
0142 4040
0143 4040
0144 0000

0145 2722 MSG14, TEXT "MR NONSTOP GAP "
0146 4016
0147 1716
0150 2324
0151 1720
0152 4007
0153 0120
0154 4040
0155 0000

0156 0213 MSG15, TEXT "BRSP SHOWN+SDWN "
0157 2320
0160 4023
0161 1004
0162 2716
0163 5323
0164 0427
0165 1640
0166 0000

0167 2205 MSG16, TEXT "READ SHUTDOWN "
0170 0104
0171 4023
0172 1025
0173 2404
0174 1727
0175 1640
0176 4040
0177 0000

0200 2722 MSG17, TEXT "WRITE XIRG "
0201 1124
0202 0540
0203 3011
0204 2207
0205 4040
0206 4040
0207 4040
0210 0000

0211 1401 MSG18, TEXT "LAST CHR TO CUR "
0212 2324

```

```

0213 4003
0214 1022
0215 4024
0216 1740
0217 0325
0220 2240
0221 0000

0222 2204 MSGT9, TEXT "RD FM BOT DELAY "
0223 4006
0224 1540
0225 0217
0226 2440
0227 0405
0230 1401
0231 3140
0232 0000

0233 2320 MSGT10, TEXT "SPCE SHOWN+SDWN "
0234 0305
0235 4023
0236 1004
0237 2716
0240 5323
0241 0427
0242 1640
0243 0000

0244 2722 MSGT11, TEXT "WRITE EOF "
0245 1124
0246 0540
0247 0517
0250 0640
0251 4040
0252 4040
0253 4040
0254 0000

0255 0522 MSGT12, TEXT "ER TO EF SP THE "
0256 4024
0257 1740
0260 0506
0261 4023
0262 2040
0263 2415
0264 0540
0265 0000

0266 2722 MSGT13, TEXT "WR TO ERASE HED "
0267 4024
0270 1740
0271 0522
0272 0123
0273 0540
0274 1005

```

```

0275 0440
0276 0000

0277 6140 MSGT14, TEXT "1 INCH DATA THE "
0300 1116
0301 0310
0302 4004
0303 0124
0304 0140
0305 2415
0306 0540
0307 0000

0310 0701 MSGT15, TEXT "GAP "
0311 2040
0312 0000

0313 4040 MSGSPG, TEXT " "
0314 4040
0315 4040
0316 4040
0317 4040
0320 4000

0321 3736 MSG556, TEXT "++556 BPI "
0322 0565
0323 0640
0324 0220
0325 1140
0326 4040
0327 4040
0330 4040
0331 4000

0332 3736 MSG200, TEXT "++200 BPI "
0333 0260
0334 0040
0335 0220
0336 1140
0337 4040
0340 4040
0341 4040
0342 4000

0343 3736 MSGEND, TEXT "++END OF TIMING-!"
0344 0516
0345 0440
0346 1706
0347 4024
0350 1115
0351 1116
0352 0737
0353 3600

```

```
0000 00000000 10100000 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 00000000 00000000 11111111
0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 00000000 00000000
0500 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000111
0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111000 00000000 00000000 00111111

1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111111 11111111 11111111 11110000 00000000 00001111
1200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1300 11111111 11111111 11111111 11111100 00000000 00000000 00000000 00000000 00000111
1400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1500 11111111 11111111 00000000 00000000 00000000 00000000 00000000 00000000 00000111
1600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1700 11111111 11110000 00000000 00000000 00000000 00000000 00000000 00000000 00000001

2000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110000 00000000
2100 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000111
2200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111110 00011111
2400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2500 11111000 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000111
2600
2700

3000
3100

3200
3300

3400
3500

3600
3700
```

```
4000
4100

4200
4300

4400
4500

4600
4700

5000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
5100 11111111 11111111 11111111 11100000 00000000 00000000 00000000 00000000 00000111
5200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
5300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111100 00011111
5400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
5500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 00000111
5600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
5700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10000000

6000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
6100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
6200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
6300 11111111 11111111 11111111 11111111 11111111 11110000 00000000 00000000 00000000
6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700
```

ADDRA	5556	GAP5CT	1707	LINK20	2346	ONESCT	0347
ALLMSG	5325	GAP6CT	1710	LINK3	0487	QTY	5416
ALLRET	5333	GAP7CT	1711	LINK50	2037	OUT	0071
ALLTBL	0160	GAP8CT	1712	LINK7	0744	OUT1	0423
ALLTU	0030	GAPCNT	2063	LINKGP	1317	OUT10	1145
ALLTU7	5304	GAPOUT	2050	LOPD	5525	OUT11	1230
ALTU79	5264	GAPS	1600	LOPRIN	5434	OUT12	1310
ARROW	5515	GAPTBL	2040	LWGR	6701	OUT13	1496
BAKSPA	5616	GETBAK	9256	M100	2504	OUT14	1511
C212	5767	HOIVND	0103	M40	5744	OUT15A	2016
C215	5766	INGH	0147	M420	2213	OUT15B	2030
C210	5765	INDX10	0010	M7	2044	OUT16	2237
C336	5770	INDX12	0012	MASK77	5743	OUT17	2265
CCAR	6704	IQTI	5024	MESSAGE	5711	OUT18	2332
CHKWRD	0240	K0001	0124	MIKECT	0107	OUT19	2360
CKUL	6726	K0100	0125	MILICT	0110	OUT2	0491
CLF	6725	K0400	0142	MK1000	5242	OUT20	2420
CLT	6712	K1000	0127	M001	6000	OUT21	2495
CNT7	0061	K1100	0122	M302	6017	OUT3	0022
CNTALL	0060	K2100	0116	M30200	6332	OUT4	0063
CRLF	5460	K212	0130	M303	6033	OUT5	0711
CWCR	6702	K215	0131	M304	6024	OUT6	0736
DATATI	2435	K240	5476	M305	6047	OUT7	1022
DECPRT	5501	K256	5477	M30550	6321	OUT8	1067
DIGIT	5564	K260	5567	M306	6096	OUT9	1117
DIVCNT	5130	K4001	0126	M307	6102	OUTGAP	1704
DIVDEN	5056	K4100	0113	M30ENG	6343	PCPU	0112
DIVIDE	5074	K4500	0117	M30SPC	6313	PRINT	5425
DIVRET	5060	K5100	0121	M30T1	6112	PRNCT	0104
DIVSOR	5127	K6100	0120	M30T10	6233	RCAR	6713
DIVSR	5057	K7100	0115	M30T11	6244	RCMR	6715
DO400	2503	K7770	0345	M30T12	6205	RDBR	6717
DOGAP2	1632	K7775	5500	M30T13	6266	READ	5633
DOGAP5	1660	K7776	0123	M30T14	6297	READ1	5651
DOGAP6	1665	K7777	0114	M30T15	6310	READ2	5670
DOGAP7	1672	KM1	0132	M30T2	6123	RESTAR	0201
DOGAP8	1677	KM10	0135	M30T3	6134	RET1	0404
DRVCNT	0342	KM13	5131	M30T4	6145	RET10	1127
DRVPT	0062	KM2	0133	M30T5	6156	RET11	1204
DRVSAV	0344	KM3	0134	M30T6	6167	RET12	1241
DRVSEL	0070	KM4	0102	M30T7	6200	RET13	1404
DRVTAB	0020	KM5	1713	M30T8	6211	RET14	1467
DVZ	5120	LCAR	6703	M30T9	6222	RET15A	2010
DV3	5110	LCMR	6705	M30T9	6222	RET15B	2021
ENDCHK	0303	LDOR	6707	M30T9	6222	RET16	2220
ENDTST	2500	LDIVND	5126	NCPU	0111	RET17	2250
ERMT12	1317	LFGR	6706	NULL	0105	RET18	2275
EXALL	5257	LINK1	0337	NUMPNT	0092	RET19	2342
EXTIME	5040	LINK11	1103	NUMTBL	0100	RET2	0434
EXTU7	5260	LINK13	1316	ODD000	0046	RET20	2403
FINI	5704	LINK15	1705	ONCE	0090	RET21	2433
GAP2CT	1706	LINK16	0144	ONES	0346	RET3	0003

MEI4	0632	T0LTU7	0084
MEI5	0673	TEMP1	0106
MEI6	0721	TEMP3	5263
MEI7	1003	TENPWR	5507
MEI8	1032	TIME	5011
MEI9	1100	TIMER	4407
MEIGAP	1604	TITLE	0067
MENCNT	5261	TRKSAV	0343
MFBR	6716	TU7	0040
RIE1	5600	TU7MSG	5345
RMBR	6714	TU7RET	5353
RWCR	6711	TU7TBL	0170
RWIND	5200	TYPECH	5731
RWIND7	5216	UPDATE	0264
RWINDA	5233	VALUE	5563
SAVANS	5073	VFS20	0077
SBMH	6727	VFSUPR	0100
SELALL	0072	VTEMP1	5565
SELCNT	5324	VTEMP2	5566
SELECT	0206	WCCA	5000
SELTU7	0073	WRICNT	5400
SET200	2462	XALL	0340
SET556	2200	XBAK	0136
SKOB	6722	XBP1	0143
SKEF	6721	XCRLF	0001
SKIP	5025	XDEC	0076
SKJD	6723	XEND	0145
SKTR	6724	XHEAD	0095
SPACE	5365	XHEAD1	0006
START	0200	XMESS	0074
T1	0400	XNULL	0132
T10	1124	XQTY	0101
T11	1200	XPRINT	0075
T12	1235	XREAD	0140
T13	1400	XREAD1	0141
T14	1463	XREAD2	0146
T15	2000	XRITE1	0137
T16	2214	XRWNO	0063
T17	2244	XRWNO7	0064
T18	2272	XTIME	0007
T19	2337	XTU7	0341
T2	0430	XWCCA	0065
T20	2400		
T21	2425		
T3	0600		
T4	0627		
T5	0670		
T6	0716		
T7	1000		
T8	1027		
T9	1074		
T0ALL	0053		

ERRORS DETECTED: 0

LINKS GENERATED: 0

RUN-TIME: 10 SECONDS

3K CORE USED

